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METHOD

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Detection and discrimination of marine oil spills and look-alike phenomena in synthetic aperture radar imagery

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ABSTRACT

Marine oil spills and several visually similar oceanic phenomena, such as biogenic slicks, low-wind areas, rain cells, and sea ice, often appear as dark patches in wide-swath synthetic aperture radar (SAR) imagery. This visual similarity among dark patches poses significant challenges for the reliable detection and discrimination of oil spills. To address this issue, we propose the Multi-class Dark Object Detection Network (MDODNet), a dedicated deep learning framework designed to simultaneously identify and classify multiple categories of dark features in wide-swath SAR images. The model is trained and validated using a cross-validated dataset comprising 11,040 annotated dark patches extracted from 2,192 Sentinel-1 VV-polarization SAR scenes. The proposed architecture enhances feature representation through an extended backbone and multi-scale feature fusion, while improving detection robustness via an anchor-based head optimized with the Focal-EIoU loss. Evaluations on an independent test dataset demonstrate strong multi-class detection performance, with average precision (AP) values of 87.04% for biogenic slicks, 93.28% for low-wind areas, 88.82% for oil spills, 90.04% for rain cells, and 89.16% for sea ice. For oil spill scenarios, detected regions are further refined using the UNet++ segmentation model to enable precise boundary delineation and quantitative area estimation. Case studies and cross-platform validation experiments further demonstrate the generalization capability and robustness of MDODNet across diverse SAR data. These results indicate that the proposed framework provides a promising solution for operational, large-scale marine oil spill monitoring.

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Oil spill; synthetic aperture radar; deep learning; detection; classification

1. Introduction

Sea surface oil spills, increasingly driven by illegal vessel discharges, pipeline leaks, natural hydrocarbon seepage, and offshore oil exploration and production, pose severe threats to marine ecosystems, coastal economies, and global environmental sustainability (Beyer et al. 2016; Dong et al. 2022). Such events can lead to water contamination, biodiversity loss, and socioeconomic disruptions, and may even escalate into transboundary environmental disputes (Kingston 2002). These impacts highlight the increasing demand for timely and reliable oil spill monitoring, in which remote sensing technologies play a critical role.

Among various Earth observation techniques, optical remote sensing systems provide rich multispectral and hyperspectral information, enabling oil type characterisation and concentration estimation (Jiao et al. 2025; Jiao, Lu, and Hu 2023; Lu et al. 2020). However, their operational effectiveness is severely constrained by dependence on solar illumination and susceptibility to cloud cover, fog, and adverse weather conditions. To overcome these limitations, spaceborne synthetic aperture radar (SAR) has become a primary tool for marine oil spill surveillance due to its all-weather, day-and-night imaging capability, high spatial resolution, and wide swath coverage (Brekke and Solberg 2005; Leifer et al. 2012). Sea surface oil spills form viscoelastic hydrocarbon layers that dampen wind-driven gravity–capillary waves, thereby reducing radar backscatter

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and appearing dark patches in SAR imagery (Caruso et al. 2013; MacDonald et al. 2015). However, similar dark patches, commonly referred to as "look-alikes", can also arise from natural oceanic phenomena, including biogenic slicks, coastal wind shadows, internal waves, low-wind areas, rain cells, sea ice, and upwelling zones (Alpers, Holt, and Zeng 2017; Najoui et al. 2018). The prevalence of these look-alike phenomena significantly complicates reliable oil spill detection and highlights the need for automated, high-precision methods capable of robust discrimination.

Early SAR-based marine oil spill detection methods primarily relied on image processing techniques and manually designed decision rules. From the late 20th to early 21st century, various image segmentation approaches were explored, including texture analysis (Marghany 2001), adaptive thresholding (Shu et al. 2010), and edge detection (Bandiera, Masciullo, and Ricci 2014), to identify dark patches in SAR imagery. These methods typically leveraged geometric and textural features of dark patches to distinguish oil spills from the surrounding seawater. Representative examples include the Bayesian-based multi-rule system (Solberg et al. 1999; Solberg, Brekke, and Husoy 2007), which integrates multiple feature thresholds to improve robustness and detection accuracy. In addition, statistical analyses of the physical and geometric characteristics of dark patches have been employed to differentiate oil spills from other oceanic look-alikes (Fiscella et al. 2010; Marghany, Cracknell, and Hashim 2009). Although these approaches are computationally efficient and relatively interpretable, they require substantial expert knowledge and extensive parameter tuning, which limits their adaptability under diverse sea states and varying SAR imaging conditions.

Polarimetric synthetic aperture radar (PolSAR) extends conventional SAR by measuring both phase and backscattering intensity across multiple polarisation channels, thereby enhancing the characterisation of the electromagnetic scattering mechanisms. This capability makes PolSAR a valuable tool for marine oil spill monitoring (Migliaccio, Gambardella, and Tranfaglia 2007; Migliaccio, Nunziata, and Buono 2015). Compared with single-polarisation SAR, fully polarimetric systems acquire the complete scattering matrix of a target, providing amplitude and phase measurements in both co-polarisation (HH, VV) and cross-polarisation (HV, VH) channels. This complementary scattering information facilitates improved discrimination between oil spills and look-alike phenomena. Early studies on oil spill detection using PolSAR data primarily relied on scattering features derived from polarimetric decomposition techniques (Migliaccio, Nunziata, and Gambardella 2009; Nunziata, Gambardella, and Migliaccio 2013; Skrunes, Brekke, and Eltoft 2014; Zhang et al. 2011). Representative methods include the Cloude–Pottier and Freeman–Durden decompositions (Cloude and Pottier 1997; Freeman and Durden 1998), which characterise distinct scattering mechanisms such as surface, double-bounce, and volume scattering in oil-covered regions (Latini, Del Frate, and Jones 2016; Li et al. 2015; Minchew, Jones, and Holt 2012). Observational analyses further indicate that clean sea surfaces and biogenic slicks typically exhibit low entropy and high anisotropy, whereas oil spills tend to produce more complex and heterogeneous scattering signatures (Skrunes et al. 2016; Skrunes, Brekke, and Eltoft 2014). Despite these advantages, the operational application of PolSAR remains constrained by the limited spatial and temporal coverage of quad-polarisation acquisition modes, as well as the sensitivity of polarimetric features to sea state variability, instrument parameters, and noise (Alpers, Holt, and Zeng 2017; Espeseth et al. 2020; Skrunes et al. 2018). These limitations complicate physical interpretation and reduce robustness, thereby motivating the development of more advanced feature extraction and denoising strategies for reliable, large-scale oil spill detection.

Compact polarimetric (CP) SAR provides enhanced scattering information compared with single-polarisation SAR, while maintaining wider swath coverage than conventional quad-polarisation imaging modes (Raney 2007; Souyris et al. 2005). In marine oil spill detection, both real and reconstructed CP SAR data have been widely used to derive compact polarimetric features, thereby improving the discrimination of oil spills from non-oil dark patches (Buono et al. 2016; Kumar et al. 2023; Li et al. 2014; Salberg and Larsen 2018; Zhang et al. 2017). However, similar to fully polarimetric SAR, CP-derived features are inherently sensitive to system noise and variability in marine conditions (Espeseth et al. 2017, 2020), which can compromise robustness under heterogeneous ocean environments. In addition, the relatively limited availability of CP SAR acquisitions constrains their scalability for large-area and long-term operational monitoring. Consequently, single-polarisation SAR data remain the most widely accessible and operationally adopted data source for marine oil pollution monitoring.

Recent advances in machine learning (ML) have significantly improved the automation of SAR-based marine oil spill detection. Traditional classifiers, including artificial neural network (ANN), random forest (RF), and support vector machine (SVM), have been widely employed to identify dark patches in SAR imagery

(Cantorna et al. 2019; Del Frate et al. 2000; Garcia-Pineda et al. 2013; Singha, Bellerby, and Trieschmann 2013; Topouzelis and Psyllos 2012; Topouzelis et al. 2007). Because individual feature often insufficient to capture the complex characteristics of dark patches, multi-feature fusion strategies have been developed that integrate geometric, textural, polarimetric, and environmental variables to improve classification performance (Garcia-Pineda et al. 2013; Guo and Zhang 2014; Kim and Jung 2018; Liu et al. 2023; Singha et al. 2016). For example, SVM combining conventional and polarimetric features has demonstrated enhanced detection accuracy (Singha et al. 2016). Despite these advances, traditional ML approaches inherently rely on manually engineered, low-level features, which limit their ability to capture broader spatial context and complex structural relationships in SAR imagery. Moreover, these methods typically evaluate features in isolation without fully accounting for the surrounding environmental context, reducing their effectiveness in discriminating oil spills from look-alike phenomena. Consequently, their generalisation capability across diverse oceanic conditions remains limited, constraining their suitability for fully automated, large-scale detection systems.

In contrast, deep learning (DL) models, particularly convolutional neural networks (CNNs), address these limitations through localised receptive fields and hierarchical convolutional operations. This architecture enables the automatic extraction of multi-scale, spatially contextualised feature representations from imagery. By jointly modelling local textures and global spatial structures, CNNs reduce the reliance on handcrafted features and substantially improve generalisation performance (He et al. 2016; LeCun, Bengio, and Hinton 2015). Early DL applications in SAR-based oil spill detection primarily focused on image-level binary classification (De Laurentiis et al. 2021; Zeng and Wang 2020). Subsequent studies adopted pixel-level semantic segmentation networks, including UNet, SegNet, and DeepLab, to delineate oil spill extents (de Moura et al. 2022; Krestenitis et al. 2019; Ma et al. 2022; Wang et al. 2024; Xie et al. 2025; Zhu et al. 2022). In parallel, object detection frameworks, including Faster R-CNN and YOLO, have been applied to localise oil spill regions in SAR imagery (Cai et al. 2024; Huang et al. 2022; Huang, Zhang, and Perrie 2025; Yekeen, Balogun, and Wan Yusof 2020). More recent approaches incorporate multi-level feature fusion and attention mechanisms to further enhance accuracy and adaptability (Fan, Sui, and Wang 2024; Song et al. 2024). Despite these advances, most DL-based methods remain limited to binary classification tasks, typically distinguishing oil spills from a single class of look-alike phenomena, such as biogenic slicks (Cui, Fan, and Zou 2025; De Laurentiis et al. 2021). In many cases, diverse look-alike phenomena are aggregated into a single non-oil category, which restricts the model's ability to simultaneously differentiate oil spills from multiple types of look-alikes (Chai et al. 2025; Jamal et al. 2025; Topouzelis et al. 2007). This limitation hinders fine-grained discrimination and reduces applicability in complex, real-world ocean environments.

Beyond oil spill detection and delineation, recent studies have increasingly focused on quantitative characterisation of marine oil pollution, particularly oil thickness estimation, concentration mapping, and volume assessment. These parameters are critical for evaluating environmental impacts, guiding response operations, and supporting damage assessment following major spill events. Multispectral and hyperspectral satellite observations have demonstrated considerable potential for oil thickness and concentration retrieval by exploiting spectral variations associated with different oil emulsification states, weathering stages, and optical properties of surface oil films (Du et al. 2025; Svejkovsky et al. 2016; Trujillo-Acatitla, Tuxpan-Vargas, and Ovando-Vazquez 2022). Meanwhile, advances in SAR remote sensing have stimulated investigations into the relationships between radar backscatter characteristics, polarisation features, damping ratios, and oil film properties (Angelliaume, Boisot, and Guérin 2018; Boisot, Angelliaume, and Guerin 2019; Garcia-Pineda et al. 2020). These studies suggest that SAR observations may provide valuable information for estimating relative oil thickness and characterising oil film conditions under all-weather and all-day observation conditions. These recent advances reflect a broader transition in marine oil spill remote sensing from qualitative detection toward quantitative characterisation of oil spill properties. Nevertheless, accurate detection and delineation of oil-contaminated regions remain a fundamental prerequisite for subsequent thickness retrieval, concentration mapping, and volume estimation. Consequently, robust discrimination between oil spills and diverse look-alike phenomena continues to represent a critical foundation for operational oil spill monitoring systems.

Distinguishing marine oil spills from diverse look-alike phenomena in wide-swath SAR imagery remains a critical challenge for operational ocean monitoring. A major bottleneck lies in the scarcity of SAR datasets with comprehensively validated annotations for multi-class dark objects, which severely constrains the

training and validation of DL models. Although several public SAR oil spill datasets have been introduced, they remain inadequate for multi-class dark object detection. For example, the M4D (Krestenitis et al. 2019), SOS (Zhu et al. 2022), and OSLM (Wang et al. 2026) datasets provide pixel-level annotations for oil spills; however, they predominantly consist of cropped image patches, restricting their applicability to wide-swath SAR imagery. While the M4D dataset also includes annotations for look-alike phenomena, it aggregates multiple look-alike types into a single category, thereby preventing fine-grained classification. A recently released Mediterranean Sea oil spill dataset provides bounding box annotations for oil spills (Yang et al. 2025), but still lacks multi-class annotations of marine dark objects. Beyond dataset limitations, mainstream DL-based object detection models, originally developed for natural images, often exhibit suboptimal performance when directly applied to SAR imagery. This performance degradation primarily arises from fundamental differences in imaging mechanisms, as well as the presence of speckle noise and complex backscatter characteristics (Zhang, Li, and Zhang 2023; Zi et al. 2024).

To overcome the aforementioned challenges, this study makes three key contributions:

(1) **Multi-class Oceanic Dark Object SAR Dataset**

A SAR dataset is constructed comprising multi-class oceanic dark objects, supported by multi-source auxiliary data to ensure annotation reliability. The dataset includes oil spills and four representative look-alike categories (biogenic slicks, low-wind areas, rain cells, and sea ice). Bounding box annotations are provided for all categories, while oil spills are additionally annotated with precise boundary masks to facilitate segmentation tasks.

(2) **Multi-class Detection Framework (MDODNet)**

The Multi-class Dark Object Detection Network (MDODNet) is proposed as a DL framework specifically designed for automated multi-class dark object detection in wide-swath SAR imagery. MDODNet enhances feature representation, incorporates multi-scale feature fusion, and employs an optimised detection head to improve robustness and detection accuracy under heterogeneous marine conditions.

(3) **Integrated Detection–Segmentation Pipeline**

To enable comprehensive oil spill monitoring, MDODNet is integrated with a UNet++ segmentation model to form a unified detection–segmentation pipeline. This framework enables both reliable multi-class dark object detection and accurate delineation of oil spill extents, thereby enhancing its operational applicability.

The remainder of this paper is organised as follows. Section 2 describes the data preprocessing procedures, the construction of the SAR dark object dataset, and the associated annotation strategy. Section 3 presents the architecture of MDODNet and its integration with the UNet++ model for oil spill delineation. Section 4 reports the evaluation results and provides a detailed discussion, including representative examples that demonstrate the generalisation capability of MDODNet across diverse SAR datasets. Section 5 concludes the paper and summarises the main findings.

2. Data preprocessing and dataset construction

2.1. Data preprocessing

As the first radar mission in the Copernicus Earth observation programme (Torres et al. 2012), the Sentinel-1 constellation currently comprises four C-band SAR sensors, Sentinel-1A, 1B, 1C, and 1D. Sentinel-1B was decommissioned in December 2021 due to a power system failure. The SAR instruments onboard these platforms provide high-resolution observations independent of weather conditions and solar illumination. Sentinel-1 primarily operates in the Interferometric Wide (IW) swath mode to produce the high-resolution Ground Range Detected (GRD) product. These products cover a swath of approximately 250 km, with a ground-range resolution of approximately 20 m $\times$ 20 m, incidence angles ranging from 31° to 46°, and noise

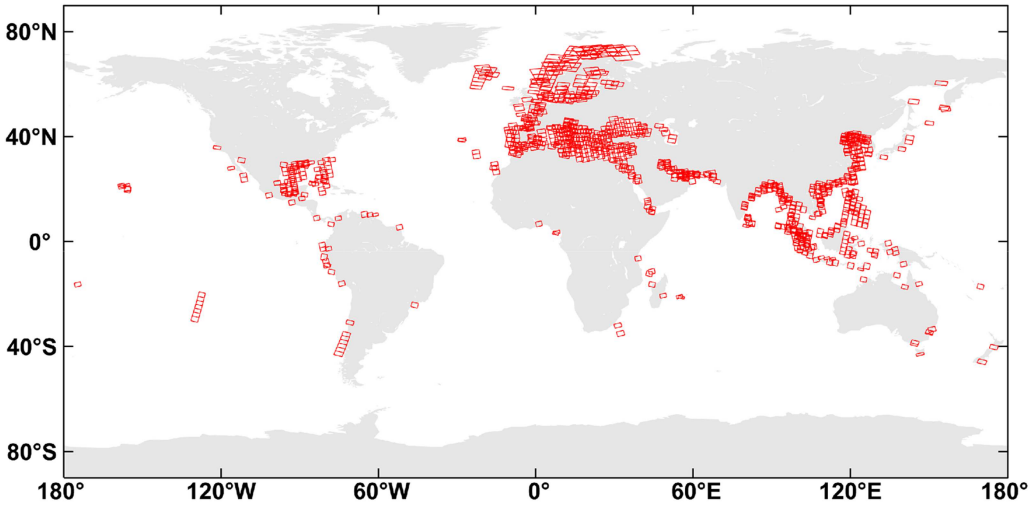


Figure 1. Global distribution of 2,192 Sentinel-1 SAR scenes (red squares) containing oil spills and visually similar look-alike phenomena, acquired between October 2014 and December 2022.

equivalent sigma zero (NESZ) values between −23 dB and −30 dB (Geudtner et al. 2014; Torres et al. 2012). The IW acquisition mode supports both single-polarisation (HH or VV) and dual-polarisation configurations (HH + HV or VV + VH). Although dual-polarisation data can provide additional information, such as improved identification of anthropogenic targets (e.g. vessels and offshore platforms), only VV-polarisation normalised radar cross section (NRCS, σ_0) was used in this study. This choice is motivated by the limited oil–water contrast typically observed in cross-polarisation channels under moderate sea states. In contrast, VV-polarisation has been shown to outperform HH-polarisation in oil spill detection, owing to its higher signal-to-noise ratio (SNR) and stronger backscatter contrast between oil-covered and clean sea surfaces (Brekke and Solberg 2005; Caruso et al. 2013). Nevertheless, the integration of dual-polarisation information represents a promising direction for future work to further enhance detection robustness and physical interpretability. A dataset comprising 2,192 Sentinel-1A/B VV-polarisation SAR imagery acquired in IW mode between October 2014 and December 2022 was compiled. Most scenes were selected from high-frequency oil spill hotspots reported in previous studies (Dong et al. 2022; MacDonald et al. 2015; Yang et al. 2025), including major shipping corridors, offshore oil production regions, and natural seep areas. Figure 1 illustrates the geographic distribution of the selected SAR images. All SAR images were manually annotated to distinguish oil spills and four representative look-alike phenomena (biogenic slicks, low-wind areas, rain cells, and sea ice). The resulting annotated dataset was subsequently used to train and evaluate MDODNet for simultaneous detection of oil spills and multiple look-alikes.

Prior to annotation, the SAR images underwent a series of preprocessing steps, including speckle noise filtering, incidence angle correction, land–sea masking, and data standardisation. Speckle noise, arising from coherent radar backscatter, degrades image quality and reduces the contrast between oil-covered and clean sea surfaces, thereby complicating the detection of dark objects. To mitigate this effect, a 10×10 pixel boxcar filter was applied, resulting in smoothed imagery with an effective pixel spacing of 100 m. Although this operation may introduce edge blurring, particularly for narrow or filamentary targets, it is considered an acceptable trade-off given the substantial reduction in feature stability caused by speckle noise. However, this smoothing may reduce the detectability of very small or low-contrast oil slicks, which will be addressed in future work through the adoption of more advanced edge-preserving filtering strategies. Land masking was implemented using the Earth Topography 2022 (ETOPO 2022) global Digital Elevation Model (DEM) to remove strong backscatter from coastal terrains that may obscure oil spill signatures (MacFerrin et al. 2025). In VV-polarisation SAR imagery, σ_0 decreases with increasing incidence angle, especially in the far-range portions of the swath, which reduces oil–water contrast and detection sensitivity. To compensate for this effect, an incidence angle correction method was applied (Wang et al. 2019), based on the empirical geophysical model function CMOD5.N (Hersbach 2010), which provides a reference factor for σ_0 recalibration. Unlike conventional implementations that assume fixed

wind conditions (e.g. wind speed of 10 m/s and relative wind direction of 45°), this study incorporates SAR viewing geometry (i.e. azimuth angle) and dynamic wind fields derived from European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis data to improve correction accuracy under varying environmental conditions. Additionally, strong backscatter from anthropogenic targets (e.g. ships, offshore platforms, buoys, and aquaculture facilities), as well as low-value pixels near image edges caused by azimuthal and range compression or initial resampling in Sentinel-1 SAR products (Ali et al. 2018), may introduce outliers. To mitigate their influence, pixel values after incidence angle correction were normalised by linearly scaling the intensity range between the 1 st and 99th percentiles to an 8-bit grayscale interval (0–255). This normalisation enhances image contrast and improves the computational efficiency of subsequent model training.

2.2. Dataset construction

To enable robust multi-class dark object detection in wide-swath SAR imagery, a cross-validated and well-annotated dataset was constructed using Sentinel-1 VV-polarisation images acquired in IW mode. This dataset consists of five categories of dark objects: marine oil spills and four representative look-alike phenomena (biogenic slicks, low-wind areas, rain cells, and sea ice). These four look-alikes were selected because they are among the commonly documented false alarms in SAR-based marine oil spill detection and can be reliably verified using auxiliary geophysical data. Comprehensive reviews of SAR oil spill monitoring have consistently identified biogenic slicks and low-wind areas as major sources of ambiguity in oil spill interpretation, while rain cells and sea ice are frequently reported as important interference factors in tropical/subtropical oceans and high-latitude environments, respectively (Alpers, Holt, and Zeng 2017; Brekke and Solberg 2005; Johansson et al. 2020; Topouzelis 2008). While other oceanographic phenomena, such as internal waves or ship wakes, may occasionally produce low radar backscatter similar to oil spills, they generally lack reliable auxiliary data sources for objective validation. Therefore, focusing on these four categories enables the dataset to capture the most representative and operationally important look-alike phenomena encountered in practical SAR-based marine oil spill monitoring. As shown in Figure 2, each oil spill or look-alike instance was individually annotated using bounding boxes, with the coordinates of the four corners recorded to support supervised model training. Because look-alike phenomena often exhibit visual characteristics similar to oil spills, manual interpretation is inherently prone to misclassification. To enhance annotation reliability and reduce labelling bias, multi-source auxiliary data are incorporated to support cross-validation during the annotation process. Representative examples of the five dark object categories, along with their corresponding auxiliary data used for verification, are presented in Figure 3.

In this study, 10 m wind speed above the sea surface, derived from the ECMWF ERA5 atmospheric reanalysis (0.25° spatial resolution, hourly temporal resolution), and precipitation data from the Global Precipitation Measurement mission (GPM) (0.1° spatial resolution, 30-minute temporal resolution) are incorporated to validate the two typical look-alike phenomena, namely low-wind areas and rain cells, observed in SAR imagery. In addition, multispectral optical imagery acquired within an eight-hour window of the corresponding SAR observations is used as the primary reference for validating biogenic slicks, oil spills, and sea ice. These optical data are obtained from multiple satellite sensors with spatial resolutions ranging from 10 m to 50 m, including the Gaofen (GF)-1/6 Wide Field of View (WFV) camera, Haiyang (HY)-1C/D Coastal Zone Imager (CZI), Huanjing (HJ)-1/2 Charge-Coupled Device (CCD), Landsat-7 Enhanced Thematic Mapper Plus (ETM+), Landsat-8/9 Operational Land Imager (OLI/OLI-2), and Sentinel-2A/B Multispectral Instrument (MSI). All optical imagery is preprocessed to generate atmospherically corrected surface reflectance products, ensuring radiometric consistency and enabling reliable cross-comparison with SAR observations.

More specifically, during the annotation process, filamentous dark patches in SAR imagery (Figure 3a) are labelled as biogenic slicks when they occur in nutrient-rich or high-chlorophyll waters, and the corresponding true-colour optical images exhibit dark or dark-grey streak-like features (Figure 3b). These features may be accompanied by surrounding green or bluish-green hues, indicative of the phytoplankton or algal activity (Gernez et al. 2023). Extensive dark regions in SAR imagery characterised by reduced radar backscatter are annotated as low-wind areas (Figure 3c) when local wind speeds derived from ERA5 are below

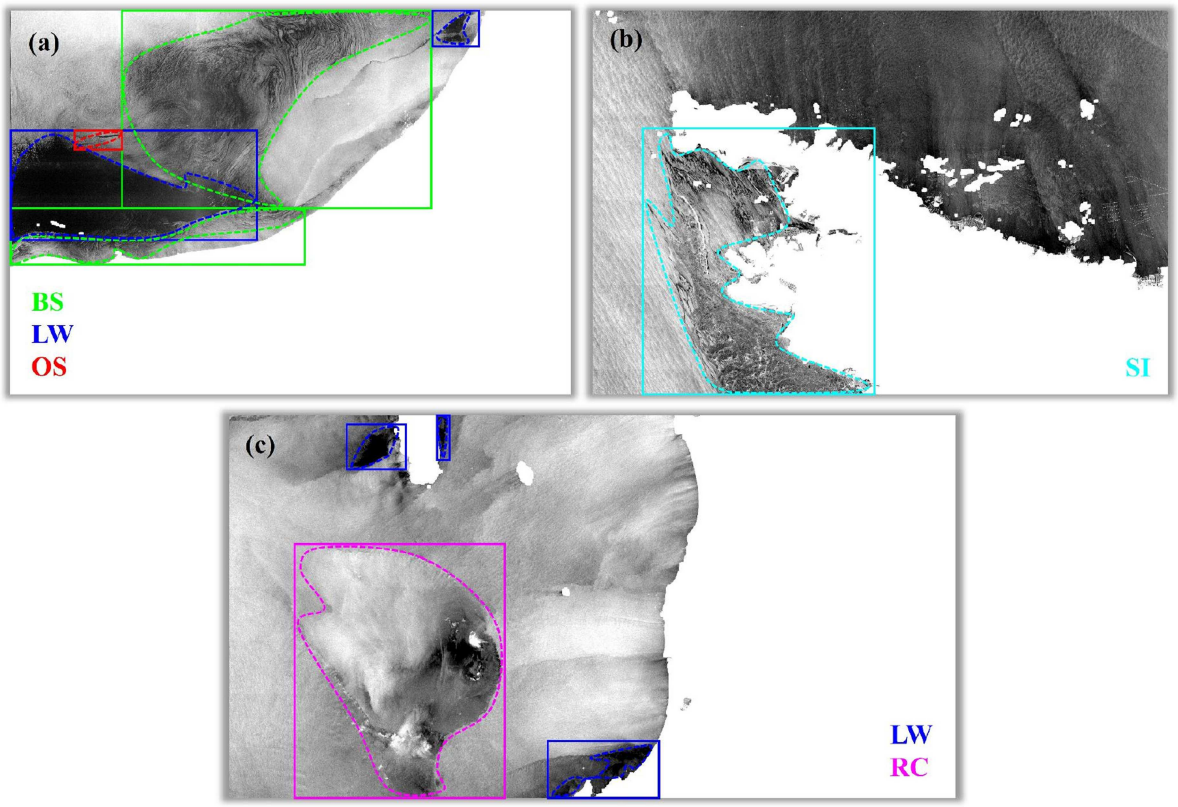


Figure 2. Schematic illustration of multi-class annotation of dark objects in SAR imagery. The green, blue, red, purple, and cyan rectangular boxes in panels (a)–(c) denote biogenic slicks (BS), low-wind areas (LW), oil spills (OS), rain cells (RC), and sea ice (SI), respectively. Dashed contours indicate the approximate spatial extent of each phenomenon.

3 m/s (Figure 3d), a condition associated with insufficient surface roughness. At low to moderate rain rates, rain-induced turbulence suppresses short Bragg waves, thereby reducing radar backscatter (Alpers et al. 2016). Accordingly, dark regions exhibiting cellular structures or alternating bright–dark patterns in SAR imagery (Figure 3g) are classified as rain cells, subject to the confirmation of active precipitation using GPM data (Figure 3h). Early-stage sea ice types, such as young ice, nilas, and grease ice, typically occur within the marginal ice zone and appear as dark patches in SAR images due to relatively low backscatter (Figure 3i). In contrast, these ice types exhibit relatively high reflectance in the visible band (Riggs, Hall, and Ackerman 1999). Therefore, dark patches in SAR images are labelled sea ice when the corresponding region in optical images appears as bright white or light grey features (Figure 3j).

Surface oil films suppress short-scale gravity–capillary waves, reducing sea surface roughness and radar backscatter, and thereby manifesting as dark patches in SAR imagery. These dark patches typically appear as elongated streaks or flocculent structures (Figure 3e), arising from processes such as illegal discharges, offshore oil production, or natural seepage. In optical imagery, the radiometric appearance of oil spills exhibits substantial variability (Lu et al. 2016), often appearing as bright, dark, or heterogeneous features containing localised bright patches associated with oil emulsification (Figure 3f). Based on these characteristics, dark patches in SAR imagery are annotated as oil spills when their spatially co-located optical signature exhibits distinctive oil spill features. In ambiguous cases where such features are not clearly identifiable, spectral signatures across visible and near-infra-red bands are further analysed. In this study, spectral profiles of marine oil spills reported by Lu et al. (2019) are used as references to distinguish oil spills from look-alike phenomena.

Auxiliary information from multiple data sources was incorporated to enhance the objectivity and accuracy of dark object annotations, thereby improving the overall quality and reliability of the dataset. In total, 2,192 Sentinel-1 VV-polarisation acquired in IW mode SAR images were collected and processed, covering diverse oceanic regions, seasons, and meteorological conditions to ensure broad spatial and

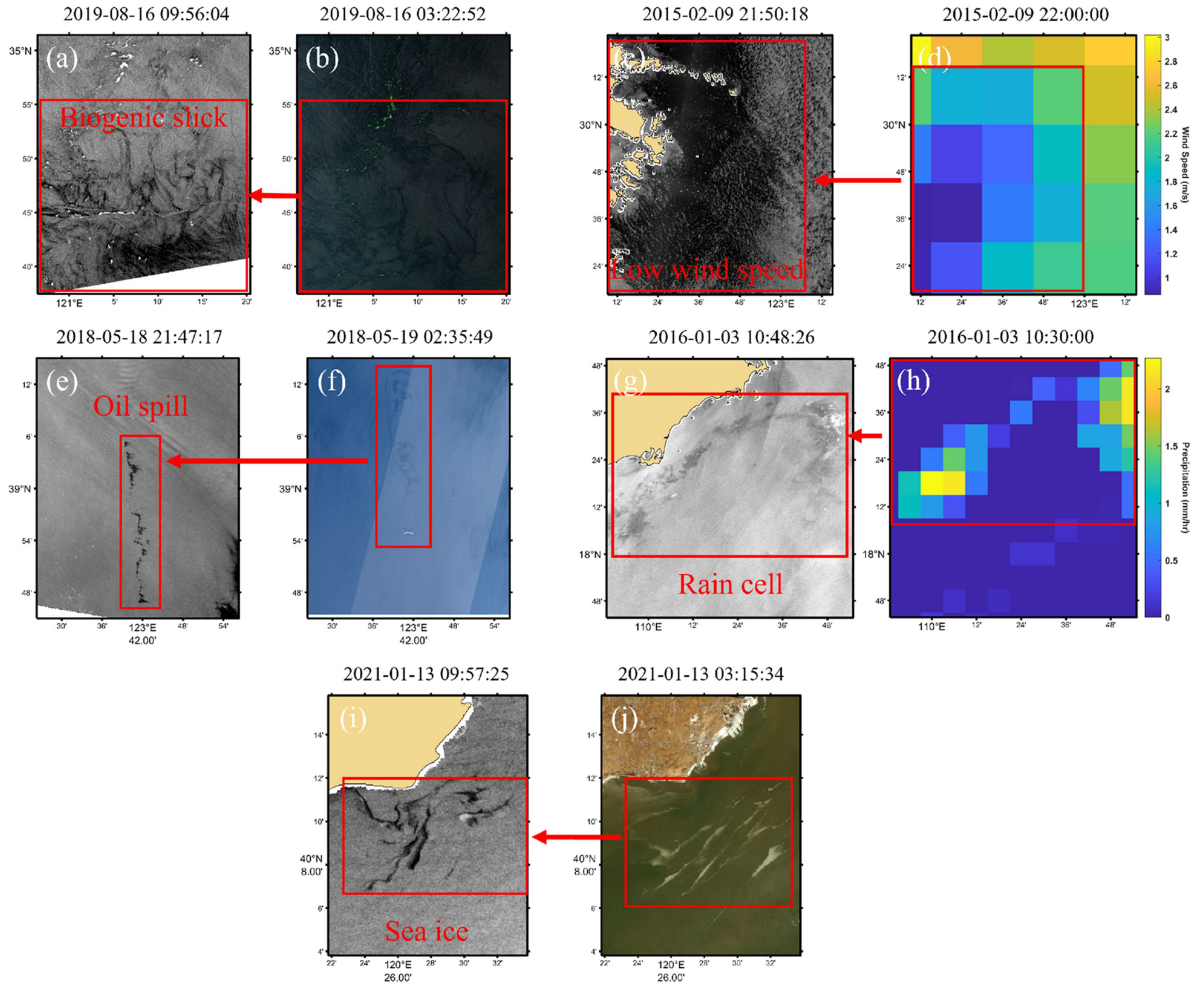


Figure 3. Representative examples of five types of dark objects in Sentinel-1 SAR imagery: biogenic slicks (a), low-wind areas (c), oil spills (e), rain cells (g), and sea ice (i). Panels (b), (f), and (j) present collocated optical imagery from GF-1, Sentinel-2B, and GF-6 corresponding to (a), (e), and (i), respectively, providing independent visual confirmation. Panel (d) shows ocean surface wind speeds derived from ECMWF reanalysis data, validating the low-wind areas identified in panel (c). Panel (h) displays rain rates from GPM measurements, with elevated precipitation values confirming the rain cells observed in panel (g).

temporal representativeness. A total of 11,040 variable-sized dark objects were manually annotated with five category labels, including 1,618 biogenic slicks, 2,002 low-wind areas, 3,218 oil spills, 2,595 rain cells, and 1,607 sea ice instances. To maintain a balanced representation at the image level, annotations for each category were distributed across 600 images, ensuring that the training process incorporates a comparable number of samples per class. The dataset was partitioned into training, validation, and test subsets with a ratio of 3:1:1. The category-wise distribution of annotated instances is summarised in Table 1. To improve model performance and generalisation while preserving the real-world spatial scale and structural characteristics of dark objects, data augmentation techniques, including rotation and flipping, were applied, increasing the dataset size by a factor of eight. More aggressive transformations (e.g. scaling, cropping, or intensity perturbation) were not adopted to avoid distorting the physical properties and morphological integrity of SAR-observed ocean features. These strategies help mitigate overfitting and enhance model robustness (Islam et al. 2024; Shorten and Khoshgoftaar 2019). Table 1 also presents the corresponding increase in annotated instances after augmentation. Additionally, pixel-level annotations were manually generated for 644 oil spill patches within the test subset to support segmentation evaluation. This subset was used to evaluate oil spill delineation performance using the UNet++ model introduced in our recent study (Huang, Zhang, and Perrie 2025).

Table 1. Distribution of (a) original and (b) augmented annotated instances across training, validation, and test datasets for five dark-object categories: biogenic slicks (BS), low-wind areas (LW), oil spills (OS), rain cells (RC), and sea ice (SI).

(a) Original annotated instances					
Subset	BS	LW	OS	RC	SI
Training	965	1,204	1,928	1,555	964
Validation	328	397	646	520	321
Test	325	401	644	520	322
Total	**1,618**	**2,002**	**3,218**	**2,595**	**1,607**
(b) After data augmentation (8)					
Subset	BS	LW	OS	RC	SI
Training	7,720	9,632	15,424	12,440	7,712
Validation	2,624	3,176	5,168	4,160	2,568
Test	2,600	3,208	5,152	4,160	2,576
Total	**12,944**	**16,016**	**25,744**	**20,760**	**12,856**

To mitigate the inherent subjectivity and potential bias in manual annotation, such as the inclusion of excessive background seawater or the omission of weak-contrast edges, a multi-expert quality control protocol was implemented. Initial bounding boxes and pixel-level masks were independently delineated and cross-validated by two senior researchers with expertise in marine remote sensing. For targets with diffuse or ambiguous boundaries, a consensus-based strategy was adopted. This multi-expert cross-validation process reduces individual subjective bias and enhances the objectivity and reliability of the ground-truth annotations.

3. Methodology

Recent advances in object detection for natural images have achieved remarkable performance, with many modern architectures pretrained using large-scale datasets such as ImageNet and COCO (Ren et al. 2017; Sohan, Sai Ram, and Rami Reddy 2024; Tian et al. 2019). However, these architectures, originally developed for natural images, often exhibit limited generalisability when directly applied to multi-class dark object detection in SAR imagery. Unlike natural images, SAR data are generated through coherent microwave backscattering and exhibit distinct statistical properties, including speckle noise, multiplicative signal fluctuations, and pronounced intensity variability. These intrinsic characteristics pose significant challenges for conventional detection networks, particularly in terms of robust feature extraction, contextual representation, and effective multi-scale feature fusion.

To address these limitations, MDODNet is developed as a deep learning architecture specifically tailored for simultaneous multi-class dark object detection in wide swath SAR imagery. The target categories include biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice. Beyond object-level detection, the proposed framework integrates a UNet++ segmentation model to enable pixel-level delineation of oil-covered regions. This unified detection–segmentation framework allows our proposed method not only to accurately identify diverse dark object types but also to generate precise oil spill boundaries at complete Sentinel-1 SAR imagery.

3.1. Multi-class dark object detection and oil spill segmentation model

Figure 4 illustrates the overall architecture of MDODNet, which comprises three main modules: a deepened backbone network, a feature-fusion neck, and an enhanced detection head. The proposed modifications are highlighted with red rectangles. The backbone is built upon the CSPDarkNet architecture adopted in the YOLO series (Sohan, Sai Ram, and Rami Reddy 2024), with increased network depth and channel width to enhance feature extraction capacity. YOLOv8 is selected as the foundational architecture for MDODNet due to its favourable balance between detection accuracy, inference efficiency, and deployment feasibility. In recent years, YOLOv8 and its variants have been widely applied in SAR object detection tasks (Cai et al. 2024; Sandalyuk and Khachatrian 2025; Yasir et al. 2024). These applications across diverse SAR scenarios further demonstrate the effectiveness and adaptability of the YOLOv8 framework.

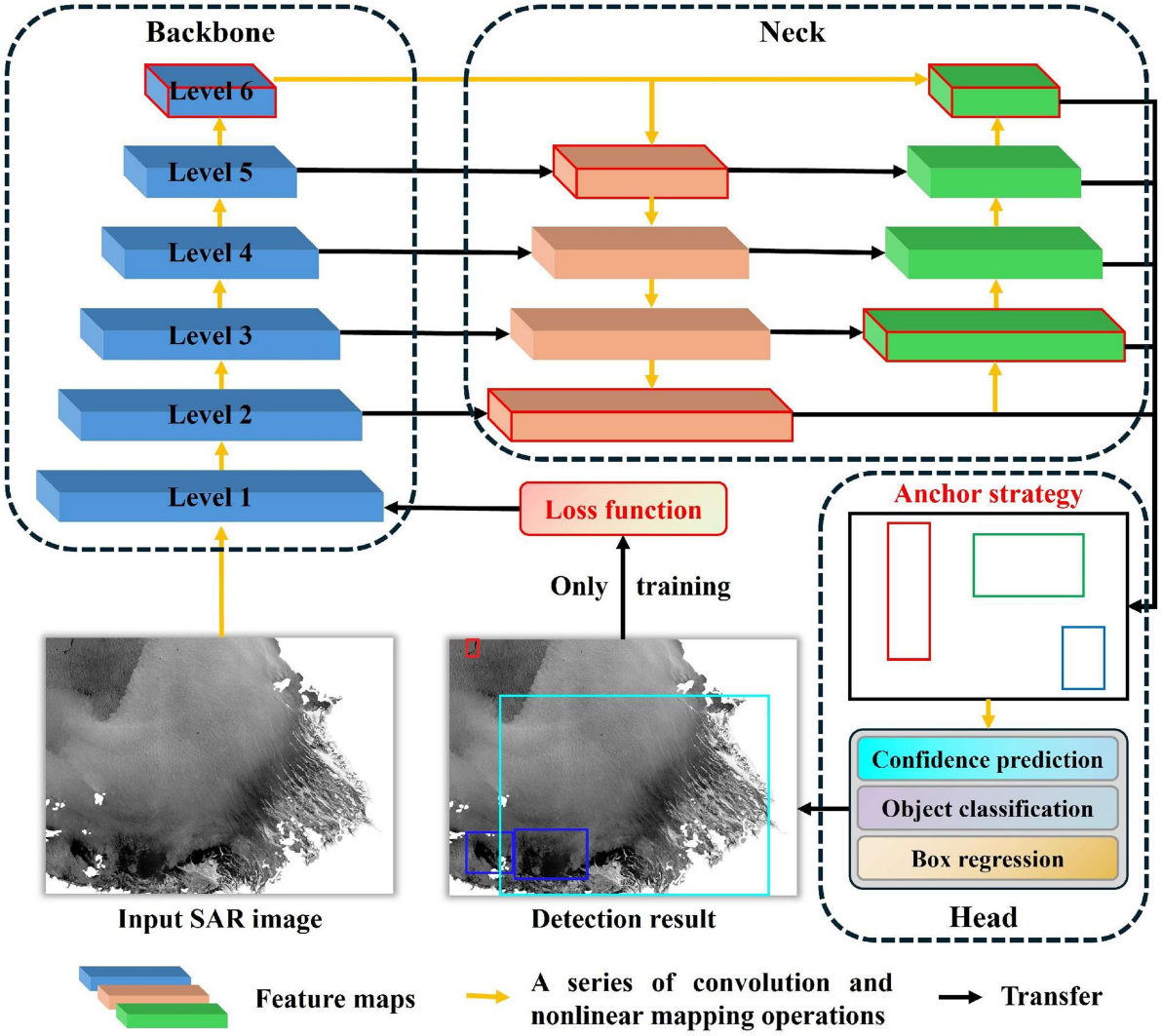


Figure 4. Overall architecture of MDODNet for simultaneous multi-class detection of oil spills and look-alike phenomena (biogenic slicks, low-wind areas, rain cells, and sea ice). The enhanced modules, including the extended backbone, expanded PAFPN, anchor-based detection head, and Focal-EIoU loss optimisation, are highlighted with red borders and annotations.

To better accommodate objects with substantial spatial variability, the original five-level feature pyramid in YOLOv8 is extended to six levels, generating hierarchical feature maps at 1/2, 1/4, 1/8, 1/16, 1/32, and 1/64 of the input image size. This extension enhances the representation of both fine-scale and large-scale dark objects. Additionally, Cross Stage Partial (CSP) modules are incorporated to expand the effective receptive field and strengthen nonlinear feature representation. These architectural enhancements enable the network to extract more discriminative semantic features in complex marine environments and to reliably detect large dark patches, which may span up to approximately 1,500 pixels (~150 km) in either spatial dimension.

In the neck module, a Path Aggregation Feature Pyramid Network (PAFPN) is incorporated to facilitate multi-scale feature aggregation across levels 2–6. By integrating a bottom-up pathway with conventional top-down pathways and lateral connections (Liu et al. 2018), PAFPN enhances bidirectional information flow, preserving fine-grained spatial details while enriching high-level semantic representations. A key innovation of MDODNet lies in its extended fusion strategy. Unlike the conventional PAFPN implementations, which typically aggregate features from levels 3 to 5, the proposed architecture integrates features from levels 2 to 6. This design is motivated by the distinct spatial characteristics of different dark object classes. Low-wind areas and sea ice often span large spatial extents, requiring deeper semantic features (e.g. level 6) for effective representation. In contrast, oil spills and biogenic slicks frequently appear as

narrow, fragmented filaments, necessitating the inclusion of shallower layers (e.g. level 2) to preserve fine-scale textural information. This extended fusion range improves multi-scale representation and contextual modelling, enabling reliable detection of both slender structures (e.g. biogenic slicks and oil spills) and large-scale patterns (e.g. low-wind areas and sea ice).

Rather than adopting an anchor-free paradigm, the detection head employs an anchor-based design (Ren et al. 2017). This choice addresses the ambiguity of object boundaries frequently encountered in complex SAR scenes. Anchor-free approaches often struggle to accurately localise overlapping or densely distributed targets with indistinct edges. In contrast, predefined anchors provide explicit scale priors, thereby stabilising bounding box regression under such conditions. Multiple detection branches operate on feature maps at different scales, with each branch specialised for objects within specific size ranges (see Table 2). Each branch jointly performs three subtasks: objectness confidence estimation, category classification, and bounding box regression.

To enhance localisation precision, the regression branch adopts the Focal and Efficient Intersection over Union (Focal-EIoU) loss (Zhang et al. 2022). Compared with conventional IoU-based losses, such as Complete IoU (CIoU) and Generalised IoU (GIoU), Focal-EIoU not only measures the spatial overlap between predicted and ground-truth bounding boxes but also introduces independent constraints for centre distance and scale discrepancies. This formulation improves bounding box regression, particularly in complex marine environments characterised by heterogeneous intensity patterns. In addition, the focal mechanism alleviates class imbalance in wide-swath SAR imagery by down-weighting abundant low-quality predictions and emphasising more informative samples. This design enhances training stability and contributes to improved overall detection performance.

During training, MDODNet is optimised using a multi-task loss function that jointly accounts for objectness prediction, category classification, and bounding box regression. These components are supervised through confidence, classification, and regression losses, respectively. The objectness prediction is optimised using a confidence loss L_{conf} based on Binary Cross-Entropy (BCE) loss:

$$L_{conf} = \frac{1}{N}\sum_{i=1}^{N}\left[g_i \log\left(\sigma\left(p_i\right)\right) + \left(1 - g_i\right)\log\left(1 - \sigma\left(p_i\right)\right)\right] \tag{1}$$

where N denotes the total number of candidate anchor boxes, g_i is the ground-truth objectness indicator for the i-th anchor, p_i is the predicted objectness score, and $\sigma()$ represents the sigmoid activation function.

The classification loss L_{cls} supervises category prediction and is also formulated using BCE to support multi-label encoding:

Table 2. Training hyperparameters and architectural settings of MDODNet for multi-class dark object detection in wide-swath SAR images.

Hyperparameter	Value
Input image size	1920×2560
Batch size	1
Number of training epochs	200
Optimiser	SGD
Initial learning rate	0.01
Momentum	0.94
Anchor sizes	(17, 10), (25, 30), (51, 76) (96, 68), (106, 122), (180, 137) (196, 128), (303, 264), (356, 542) (436, 615), (739, 706), (925, 792) (1116, 947), (1309, 1291), (1553, 2097)
IoU threshold (training)	0.7
IoU threshold (detection)	0.5
NMS threshold (training)	0.7
NMS threshold (detection)	0.5
Maximum detection per image	100
Focal-EIoU loss modulation factor (γ)	0.5
Class weights (BS, LW, OS, RC, SI)	2.0, 1.6, 1.0, 1.2, 2.0
Confidence loss weight (λ_{conf})	1
Classification loss weight (λ_{cls})	1
Regression loss weight (λ_{reg})	0.5

$$L_{cls} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} w_j\left[g_{ij}\log\left(\sigma\left(p_{ij}\right)\right) + \left(1 - g_{ij}\right)\log\left(1 - \sigma\left(p_{ij}\right)\right)\right] \tag{2}$$

where C is the number of object classes, g_{ij} indicates whether the i -th anchor box belongs to class j, p_{ij} is the predicted probability for class j, and w_j is the class-specific weight. To mitigate class imbalance, category-dependent weights are assigned based on the inverse frequency of annotated instances. Specifically, the weights for biogenic slicks (BS), low-wind areas (LW), oil spills (OS), rain cells (RC), and sea ice (SI) are set to 2.0, 1.6, 1.0, 1.2, and 2.0, respectively. To mitigate class imbalance, category-dependent weights are assigned based on inverse frequency. Specifically, weights are computed relative to the majority class (oil spills), thereby increasing the contribution of underrepresented categories during training.

The regression loss L_{reg} adopts the Focal-EIoU formulation, which incorporates geometric constraints and a focal modulation factor:

$$L_{reg} = \frac{1}{N}\sum_{i=1}^{N} IoU_i^{\gamma}\left[1 - IoU_i + \frac{\rho^2\left(p_i, p^{gt}\right)}{\left(w_i^c\right)^2 + \left(h_i^c\right)^2} + \frac{\rho^2\left(w_i, w^{gt}\right)}{\left(w_i^c\right)^2} + \frac{\rho^2\left(h_i, h^{gt}\right)}{\left(h_i^c\right)^2}\right] \tag{3}$$

where γ is the focal modulation factor (set to 0.5), IoU_i denotes the overlap between the predicted and ground-truth bounding box, and $\rho()$ represents the Euclidean distance. Here, p_i and p^{gt} are the central coordinates of the predicted and ground-truth boxes, respectively; w_i, h_i and w^{gt}, h^{gt} denote their widths and heights; and w_i^c, h_i^c correspond to the width and height of the smallest enclosing box. The focal modulation term IoU_i^{γ} assigns higher weights to well-localised predictions, encouraging precise bounding box regression and improving convergence stability in complex SAR scenes.

The overall loss function L_{total} is defined as:

$$L_{total} = \lambda_{conf}L_{conf} + \lambda_{cls}L_{cls} + \lambda_{reg}L_{reg} \tag{4}$$

where λ_{conf}, λ_{cls}, and λ_{reg} are weighting coefficients for the respective loss terms. In this study, these are empirically set to 1, 1, and 0.5, respectively. Additional training hyperparameters are summarised in Table 2. The input image size, anchor box dimensions, and class weights are determined based on statistical analysis of the constructed dataset, while the remaining parameters follow commonly adopted configurations in related studies.

Following multi-class dark object detection, pixel-level segmentation of oil spill regions is performed to quantify their spatial extent and morphological characteristics. While object detection provides localisation of potential spills, accurate estimation of spill area, shape, and boundary complexity requires fine-grained segmentation. To this end, the UNet++ semantic segmentation model is employed, which has demonstrated strong capability in accurately delineating oil slick boundaries from SAR observations (Huang, Zhang, and Perrie 2025). Compared with end-to-end instance segmentation frameworks, this decoupled detection–segmentation strategy significantly reduces manual annotation costs while maintaining high spatial precision in boundary delineation. Nevertheless, exploring unified instance segmentation architectures remains an important direction for future work. Specifically, SAR image patches corresponding to oil spill regions identified by MDODNet are extracted and fed into UNet++, which generates binary masks for precise boundary delineation. By incorporating dense skip connections and multi-scale decoder pathways, UNet++ produces enriched feature representations and facilitates more accurate delineation of weak, diffuse, or irregular boundaries (Zhou et al. 2020). This capability is particularly advantageous for oil spills exhibiting blurred edges and heterogeneous textural patterns in SAR imagery. As the architecture and training strategy of UNet++ have been comprehensively described and validated in our previous work, only a brief overview is provided here. The resulting segmentation outputs are subsequently used to estimate spill area quantitatively, forming the basis for downstream evaluation and validation.

3.2. Performance metrics

Model performance for multi-class dark object detection and oil spill segmentation is evaluated using four metrics: IoU, Precision, Recall, and Average Precision (AP). All metrics are bound within [0, 1], with larger values indicating better performance.

IoU measures the spatial overlap between predictions and their corresponding ground-truth.

$$IoU = \frac{TP}{TP + FP + FN} \tag{5}$$

where *TP*, *FP*, and *FN* denote the number of true positives (correct detections), false positives (incorrect detections), and false negatives (missed detections), respectively. Higher IoU values indicate more accurate predictions.

Precision is defined as the proportion of correct detections among all predicted positives.

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

High precision indicates strong reliability with a low false-alarm rate.

Recall quantifies the proportion of correct detections to the total number of ground-truth samples.

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

A high Recall indicates that most true targets are successfully detected.

AP provides a comprehensive evaluation by summarising the Precision–Recall trade-off across multiple confidence thresholds:

$$AP = \sum_{n=1}^{N} (R_n - R_{n-1}) P_n \tag{8}$$

where N is the number of confidence thresholds, and P_n and R_n denote Precision and Recall at the n-th threshold, respectively. A higher AP indicates a better balance between Precision and Recall over varying confidence levels.

For multi-class dark object detection, the performance of the MDODNet is evaluated using Precision, Recall, and AP. For oil spill segmentation using UNet++, Precision, Recall, and IoU are employed to assess pixel-level delineation accuracy.

4. Experimental results and analysis

4.1. Quantitative evaluation of multi-class dark object detection

To evaluate the effectiveness of MDODNet for multi-class dark object detection in SAR imagery, extensive training and validation experiments were conducted using the annotated dataset. Figure 5 illustrates the evolution of training and validation loss values throughout the learning process, where each epoch corresponds to a complete pass through the training dataset. The loss quantifies the discrepancy between model predictions and ground-truth annotations, and its temporal evolution reflects the learning dynamics and convergence behaviour of the network.

A rapid decrease in loss is observed during the early training stage (Figure 5), indicating that MDODNet effectively learns discriminative features for different dark object categories. In subsequent epochs, both training and validation losses gradually stabilise, with only a small gap between the two curves, suggesting stable optimisation without evident overfitting. The validation loss converges after approximately 60 epochs, and the absence of significant oscillations further indicates robust convergence. The minimum validation loss (0.091) is achieved at epoch 177, and the corresponding model checkpoint is selected for subsequent detection experiments.

The detection performance of MDODNet is further evaluated using Precision–Recall (PR) curves for all five dark object categories, as shown in Figure 6. PR curves characterise the trade-off between Precision and Recall across varying confidence thresholds. An effective detector should achieve high values for both metrics; however, in practice, improvements in one often occur at the expense of the other. Consequently, selecting appropriate confidence thresholds is essential for achieving balanced detection performance. In this study, class-specific confidence thresholds are determined by optimising the balance between Precision and Recall for each category. The optimal thresholds are set to 0.27 for biogenic slicks, 0.38 for

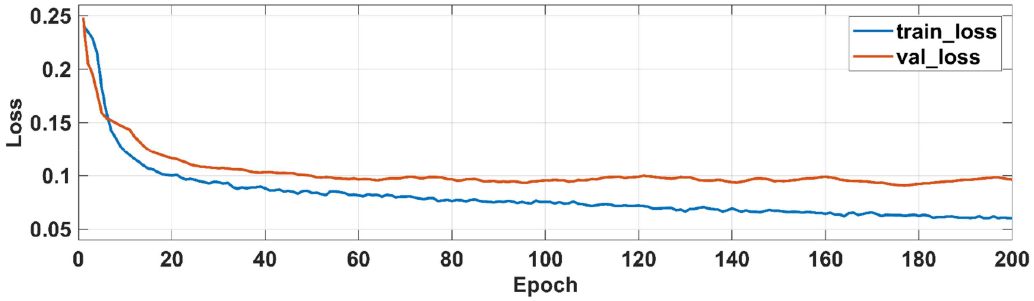


Figure 5. Training and validation losses over epochs for the MDODNet model, illustrating the optimisation process for simultaneous detection of oil spills and look-alike phenomena (biogenic slicks, low-wind areas, rain cells, and sea ice).

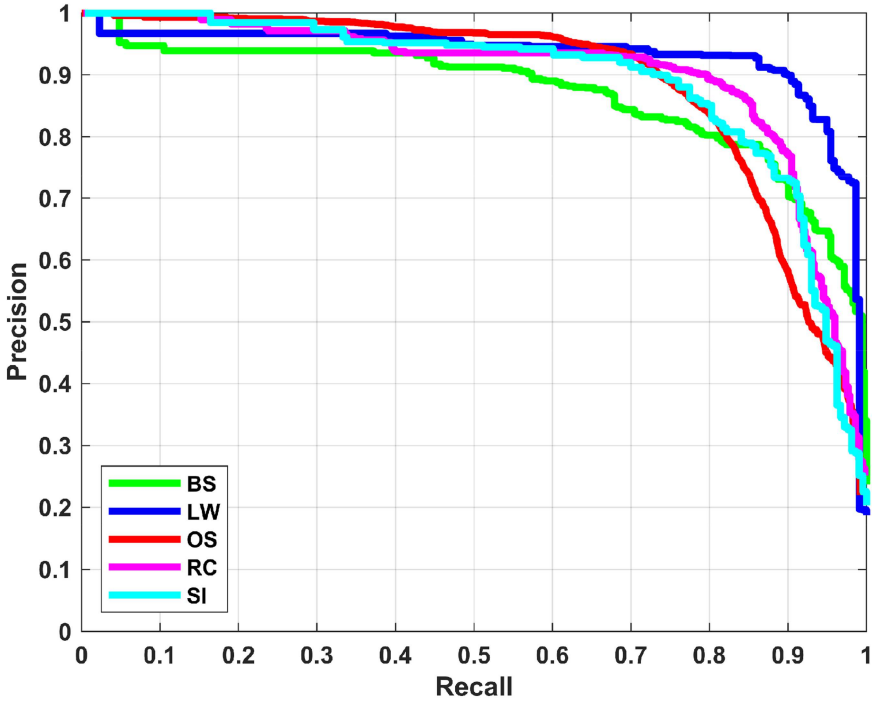


Figure 6. Precision–Recall (P–R) curves of MDODNet on the test dataset, illustrating multi-class detection performance for oil spills (OS), biogenic slicks (BS), low-wind areas (LW), rain cells (RC), and sea ice (SI).

low-wind areas, 0.25 for oil spills, 0.36 for rain cells, and 0.26 for sea ice. These thresholds were subsequently applied to all quantitative evaluation experiments.

Quantitative performance was evaluated on the test dataset using Precision, Recall, and AP, with results summarised in Experiment I of Table 3. For oil spill detection, MDODNet achieved Precision, Recall, and AP values of 81.49%, 81.42%, and 88.82%, respectively, demonstrating its effectiveness in discriminating oil slicks from visually similar dark ocean features. Although oil spills are the primary focus of this study, they do not exhibit the highest detection performance among the five categories. The AP values, ranked from highest to lowest, are as follows: low-wind areas (93.28%), rain cells (90.04%), sea ice (89.16%), oil spills (88.82%), and biogenic slicks (87.04%). These performance variations are likely attributable to differences in background conditions and the spatial and textural complexity of each object category. Low-wind areas typically appear as large, homogeneous dark regions with smooth boundaries, providing consistent structural cues that facilitate robust feature learning. Rain cells, despite moderate texture variability, often exhibit circular or elliptical patterns with relatively well-defined boundaries, enabling reliable detection. Sea ice generally forms dense clusters with distinct macroscopic boundaries, offering strong geometric

Table 3. Comparative evaluation of multi-class dark object detection using MDODNet under three experimental settings.

Experiment I: Five-class fine-grained detection					
Metric	OS	BS	LW	RC	SI
Precision (%)	81.49	80.14	89.85	85.37	81.55
Recall (%)	81.42	80.38	89.96	85.26	81.68
AP (%)	88.82	87.04	93.28	90.04	89.16
Experiment II: Binary detection (Oil vs. Non-oil)					
Metric				OS	Non-oil (BS + LW + RC + SI)
Precision (%)				80.81	82.79
Recall (%)				80.78	82.79
AP (%)				87.99	88.67
Experiment III: Oil spill detection only					
Metric				OS	
Precision (%)				78.69	
Recall (%)				78.63	
AP (%)				85.63	

Experiment I: Fine-grained five-class detection of oil spills (OS), biogenic slicks (BS), low-wind areas (LW), rain cells (RC), sea ice (SI).

Experiment II: Binary detection, in which the four look-alike phenomena (BS, LW, RC, and SI) are merged into a single non-oil category and distinguished from oil spills.

Experiment III: Oil spill detection only, where look-alike phenomena are excluded from both training and testing.

features for classification. In contrast, oil spills exhibit greater morphological variability, influenced by sea state, wind conditions, and imaging geometry. Some oil spills present diffuse boundaries or limited spatial extents, which increases the difficulty of robust feature extraction. Biogenic slicks pose the greatest challenge, as they are characterised by irregular shapes, fragmented distributions, diffuse edges, and weak contrast relative to surrounding seawater. These characteristics provide limited stable discriminative features, thereby reducing detection performance in complex marine environments.

For the oil spill segmentation task, the previously validated UNet++ model was employed to perform pixel-level delineation of oil spill regions identified by MDODNet (Huang, Zhang, and Perrie 2025). A comprehensive evaluation was conducted using the constructed oil spill segmentation dataset. The segmentation results demonstrate strong performance, with Precision, Recall, and IoU values of 89.43%, 89.51%, and 80.90%, respectively. These results indicate that UNet++ accurately delineates oil spill boundaries with high spatial fidelity. The close agreement between Precision and Recall further suggests balanced segmentation performance, indicating robust delineation capability across diverse sea states and varying spill morphologies.

Efficient localisation and segmentation of oil spills are essential for timely marine monitoring and for supporting rapid response and cleanup operations. The runtime performance of the proposed framework was evaluated on a computing platform equipped with an Intel Core i9-10900X CPU, an NVIDIA GeForce RTX 3090 GPU (24 GB VRAM), and 32 GB of system RAM. MDODNet requires approximately 260 billion floating-point operations (FLOPs) and contains 68.3 million parameters, indicating a computationally efficient architecture suitable for inference on standard GPU platforms. The peak GPU memory consumption per SAR scene is approximately 10 GB. The preprocessing pipeline processes a full-resolution SAR image in approximately 50 s, while multi-class dark object detection requires approximately 0.8 s per image and segmentation of an individual oil spill patch requires about 0.07 s. Although the total runtime depends on image size and the number of detected oil spill regions, processing a full-size SAR scene containing up to 10 oil spills can be completed within 1 min. These results indicate that the proposed detection–segmentation framework achieves high computational efficiency and is capable of supporting near-real-time monitoring of oil spill events using SAR observations.

4.2. Comparative analysis of different detection strategies

To assess the impact of fine-grained classification on multi-class dark object detection in SAR imagery, three comparative experiments were conducted using the same annotated dataset for both training and testing. The experimental configurations are summarised as follows:

- **Experiment I:** Fine-grained five-class detection, including biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice (see Section 4.1).
- **Experiment II:** Binary detection, in which the four look-alike categories (biogenic slicks, low-wind areas, rain cells, and sea ice) are merged into a single non-oil category and distinguished from oil spills.
- **Experiment III:** Oil spill-only detection, where all look-alike phenomena are excluded from both training and testing.

As summarised in Table 3, Experiment I achieves the highest AP for oil spill detection (88.82%), outperforming Experiment II (87.99%) and Experiment III (85.63%). Compared with Experiment I, Experiment II shows a decrease of 0.83% in oil spill AP, and its AP for look-alike detection is 1.21% lower than the mean AP of four look-alike classes in Experiment I. These results indicate that, although Experiment II can effectively distinguish oil spills from non-oil objects, it is less capable of capturing the subtle structural and textural differences among distinct look-alike categories.

Experiment III yielded the lowest performance, with Precision, Recall, and AP values of 78.69%, 78.63%, and 85.63%, respectively. Its reduced effectiveness is particularly evident in complex SAR scenes containing multiple types of dark objects. This degradation can be attributed to training exclusively on oil spill samples, which constrains the model's ability to distinguish oil spills from visually similar non-oil dark patches. Similarly, aggregating heterogeneous look-alike phenomena into a single class in Experiment II neglects their structural, textural, and contextual diversity, thereby weakening discriminative feature learning.

In contrast, the fine-grained detection strategy in Experiment I enables MDODNet to learn category-specific characteristics, including texture distribution, boundary morphology, intensity variation, and contextual cues. This refined feature representation improves discrimination among dark object classes, reduces misclassification, and enhances detection robustness and generalisation. Overall, these findings demonstrate that fine-grained classification not only improves detection accuracy but also strengthens semantic interpretability and operational reliability, which are critical for effective marine oil spill monitoring and emergency response.

4.3. Contribution analysis of key modules

A comprehensive ablation study was conducted to evaluate the contribution of each enhanced module in MDODNet. As MDODNet is developed as an improved variant of YOLOv8, the original YOLOv8 architecture was adopted as the baseline. Each proposed module was incrementally integrated into the baseline, and the performance was evaluated under identical training and testing conditions to ensure a fair comparison. This controlled experimental design enables isolation of the individual impact of each module, as well as assessment of their cumulative contributions to overall detection performance.

Specifically, MDODNet introduces three key enhancements:

(1) **Deep Feature Fusion (DFF) via extended backbone and PAFPN structure:**
The backbone depth is extended from 5 to 6 levels, and the PAFPN fusion range is expanded from levels 3–5 to 2–6, enabling richer multi-scale feature aggregation. This design strengthens cross-scale semantic and spatial information integration, thereby improving detection of both small-scale and large-scale dark objects.

(2) **Anchor-based detection head**
The original anchor-free head in YOLOv8 is replaced with an anchor-based detection mechanism. This modification improves the stability and accuracy of bounding box regression, particularly for irregularly shaped or scale-variable targets.

(3) **Focal-EIoU loss function**
The Focal-EIoU loss replaces the standard box regression loss, promoting faster convergence, alleviating sample imbalance, and improving localisation accuracy by jointly optimising overlap ratio, centre distance, and aspect ratio alignment.

Through the progressive integration of these modules, the ablation results provide a clear understanding of how architectural refinement, detection head redesign, and loss function optimisation collectively contribute to the performance improvements achieved by MDODNet.

To ensure a fair comparison, all models were trained and evaluated using the same dataset and experimental settings, with results summarised in Table 4. Each proposed enhancement contributes positively to detection performance, with the DFF module yielding the most significant improvement. Specifically, the AP for oil spill detection increases by 3.79%, while AP values for the four look-alike categories, biogenic slicks, low-wind areas, rain cells, and sea ice, improve by 0.99%–2.74%. These consistent gains indicate that enhanced multi-scale feature aggregation benefits both oil spill detection and discrimination among different look-alike phenomena.

Unlike the conventional PAFPN in YOLOv8, which primarily fuses intermediate feature levels (levels 3–5), the proposed extended fusion strategy incorporates feature maps from levels 2–6. By integrating shallower layers rich in fine-grained spatial details with deeper layers containing high-level semantic information, the DFF module strengthens cross-scale representation capability. This design improves sensitivity to small-scale targets, such as fragmented oil spills, while simultaneously enhancing contextual awareness for large, spatially extensive dark patches, such as low-wind areas. Consequently, the extended fusion architecture improves both localisation precision and class discrimination robustness across heterogeneous object scales.

Replacing the anchor-free detection head with an anchor-based design further enhances detection robustness, particularly in scenes containing densely distributed or irregularly shaped dark patches such as oil spills, biogenic slicks, and rain cells. Anchor-free detectors typically regress a single bounding box per spatial location, which limits their ability to represent overlapping, adjacent, or partially occluded targets. This limitation may lead to missed detection or suboptimal bounding box fitting in complex scenarios where multiple dark objects occur in close proximity.

In contrast, the anchor-based strategy introduces multiple predefined anchor priors at each spatial location, enabling the model to regress multiple candidates bounding boxes with diverse aspect ratios and scales. This mechanism improves localisation flexibility and fitting accuracy for heterogeneous dark object morphologies. Quantitatively, the anchor-based modification increases the AP for oil spill detection by 2.40% and yields additional gains of 2.15%–3.16% across the remaining categories. These results demonstrate that anchor priors provide structural advantages in complex marine environments characterised by scale variability, shape irregularity, and spatial clustering.

Finally, incorporating the Focal-EIoU loss function results in consistent, albeit relatively modest, performance improvements. The AP increases by 0.11%–0.77% across all categories, while the training process converges approximately 30 epochs earlier compared with the baseline loss configuration. By introducing a focal modulation term, Focal-EIoU down-weights low-quality or ambiguous predictions and emphasises high-quality samples during optimisation. This mechanism is particularly beneficial for wide-swath SAR imagery, where dense candidate regions and complex backscatter variability can otherwise lead to numerous low-quality proposals and unstable regression updates. In addition, explicit geometric constraints on overlap, centre distance, and aspect ratio further improve localisation consistency.

To provide a more comprehensive evaluation of MDODNet, its performance was compared with several representative object detection models, including SSD (Liu et al. 2016), FCOS (Tian et al. 2019), and YOLOX (Ge et al. 2021). Although these models were originally developed for natural-image applications, they have been widely adopted in remote sensing and SAR target detection studies, making them strong benchmark models for comparison. In addition, there are currently few publicly available deep learning frameworks specifically designed for multi-class marine dark object detection in wide-swath SAR imagery. To ensure a rigorous and fair comparison, all models were trained and evaluated on the same dataset, data

Table 4. Ablation study results for multi-class dark object detection (AP, %), comparing the YOLOv8 baseline with MDODNet under incremental integration of architectural enhancements. BS, LW, OS, and SI denote biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice, respectively.

Model	BS	LW	OS	RC	SI
Baseline (YOLOv8)	83.73	89.42	82.02	83.27	84.73
Baseline + DFF	84.72	90.98	85.81	86.01	86.50
Baseline + DFF + Anchor	86.87	93.17	88.21	89.27	88.86
Baseline + DFF + Anchor + Focal-EIoU (MDODNet)	87.04	93.28	88.82	90.04	89.16

augmentation strategies, optimisation settings, and hardware configurations. The quantitative comparison results are summarised in Table 5. MDODNet consistently achieves the highest AP across the five dark object categories. The results indicate that, although the benchmark models provide reasonable performance, they are less effective in capturing the complex structural, textural, and contextual characteristics of marine oil spills and diverse look-alike phenomena in SAR imagery. These findings further demonstrate the effectiveness of the proposed architectural enhancements and highlight the suitability of MDODNet for multi-class marine dark object detection in SAR-based oil spill monitoring applications.

In summary, the three targeted enhancements in MDODNet, extended deep feature fusion, anchor-based detection, and Focal-EIoU loss optimisation, jointly contribute to measurable performance gains in multi-class dark object detection. The ablation study systematically validates the individual and combined effectiveness of these modules. These improvements enhance detection robustness across heterogeneous object scales and morphologies, highlighting the practical potential of MDODNet for large-scale, operational marine monitoring using SAR imagery.

4.4. Representative case study

To further evaluate the capability of MDODNet in detecting multiple types of dark objects under complex marine conditions, three representative case studies were analysed. Figure 7a presents the detection results for a Sentinel-1 VV-polarisation SAR scene acquired in IW mode over the Taiwan Strait at 10:01 UTC on June 11, 2021. In this scene, MDODNet successfully identified one oil spill (red bounding box), three biogenic slicks (green bounding boxes), and one low-wind area (blue bounding box), with IoU values of 0.91, 0.72–0.92, and 0.86, respectively, relative to the ground truth. The predicted confidence scores vary across the detected objects, reflecting differences in background heterogeneity, morphological complexity, local contrast, and signal-to-noise conditions. The simultaneous presence of multiple dark object categories with distinct spatial scales and geometric characteristics highlights the complexity of marine environments. Notably, the detected oil spill exhibits an elongated, irregular morphology, consistent with linear discharge patterns commonly observed in high-traffic maritime corridors. Given the dense shipping activity in the Taiwan Strait, such strip-like features are likely associated with anthropogenic discharge events, although in situ observations are required for definitive confirmation. These results demonstrate that MDODNet can effectively discriminate among multiple classes of dark objects within a single scene, even when they coexist and exhibit partially overlapping spatial distributions.

To further validate the detection results, multiple auxiliary datasets were incorporated for cross-verification. Figure 7b presents a GF-1 optical image acquired at 02:31 UTC on the same day, covering a nearby portion of the SAR scene shown in Figure 7a. Despite an approximately 8-hour temporal offset, a strip-like dark patch with distinct white fringes, characteristic of emulsified oil (Lu et al. 2020), is clearly visible in the optical imagery. The spatial location and morphological characteristics of this feature are consistent with the oil slick identified in the SAR observation, suggesting that both observations correspond to the same pollution event. The three biogenic slick regions detected by MDODNet are located in nutrient-rich nearshore waters and exhibit flocculent, irregular patterns typical of natural surface films. Importantly, no oil-related spectral signatures or discoloration are observed in the corresponding optical image, further supporting their classification as biogenic slicks rather than anthropogenic oil spills.

Figure 7c presents ECMWF ERA5 reanalysis wind fields at 10:00 UTC on June 11, 2021. A localised region with wind speeds below 3 m/s is evident on the western side of the study area, spatially coinciding with the low-wind patch detected by MDODNet. This spatial agreement confirms that the detected feature is

Table 5. Quantitative performance comparison (AP, %) of MDODNet and three representative object detection models (SSD, FCOS, and YOLOX) for multi-class marine dark object detection in SAR imagery. BS, LW, OS, RC, and SI denote biogenic slick, low-wind area, oil spill, rain cell, and sea ice, respectively.

Model	BS	LW	OS	RC	SI
SSD	74.79	76.77	74.43	76.68	78.00
FCOS	84.52	89.09	80.90	82.50	85.87
YOLOX	83.47	88.31	81.04	82.58	84.55
MDODNet	87.04	93.28	88.82	90.04	89.16

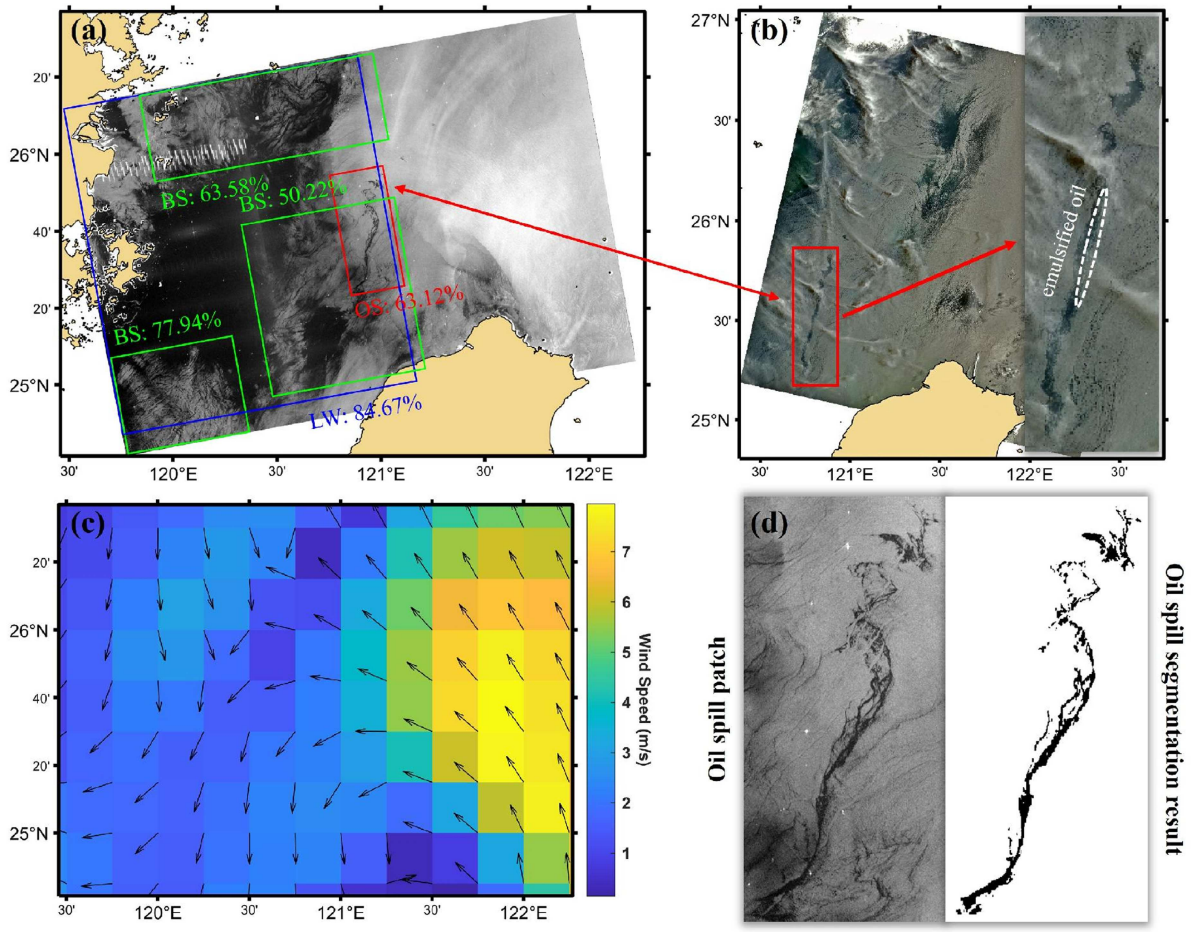


Figure 7. (a) Multi-class dark object detection results obtained using MDODNet from a Sentinel-1A VV-polarisation SAR image acquired in interferometric Wide (IW) mode at 10:01 UTC on June 11, 2021. Green, blue, and red rectangular boxes denote correctly detected biogenic slicks (BS), low-wind areas (LW), and oil spills (OS), respectively. Numerical values indicate the detection confidence scores. (b) GF-1 Wide Field of View (WFV) true-colour composite image acquired at 02:31 UTC on June 11, 2021, covering the area shown in panel (a). The white elliptical shape highlights the emulsified oil spill observed in the optical imagery. (c) ECMWF ERA5 reanalysis wind field at 10:00 UTC on June 11, 2021. (d) Cropped oil spill region detected by MDODNet and its corresponding segmentation result generated by the UNet++ model.

associated with reduced surface roughness and, consequently, diminished radar backscatter under calm wind conditions.

Figure 7d presents the pixel-level segmentation of the detected oil spill using UNet++, achieving a precision of 91.31% and a recall of 89.69%. The segmentation result accurately delineates the oil spill boundary while effectively avoiding misclassification of the adjacent biogenic slicks, yielding an estimated oil spill area of 63.80 km^2. This case study highlights the complementary roles of MDODNet in object-level identification and UNet++ for accurate boundary delineation, enabling reliable localisation and quantitative assessment of oil slicks in complex marine environments where multiple types of dark objects coexist.

Figure 8a presents the detection results of MDODNet applied to a Sentinel-1B VV-polarisation SAR scene acquired in IW mode at 21:47 UTC on January 18, 2018, over Korean Bay, where floating ice and ice–water mixtures are prevalent during winter. In this scene, the model successfully identifies two low-wind areas (blue bounding boxes) with IoU values of 0.85–0.90 and two sea ice regions (cyan bounding boxes) with IoU values of 0.78–0.88, demonstrating its capability to discriminate among multiple dark object types in a complex cryospheric marine environment. Several sea ice regions exhibit dark patches in the SAR imagery. This backscatter reduction is likely associated with grease ice, newly formed thin ice, or fragmented floating ice mixed with seawater. These surface conditions suppress gravity–capillary waves and reduce surface roughness at radar wavelength scales, resulting in reduced radar backscatter. Such signatures are

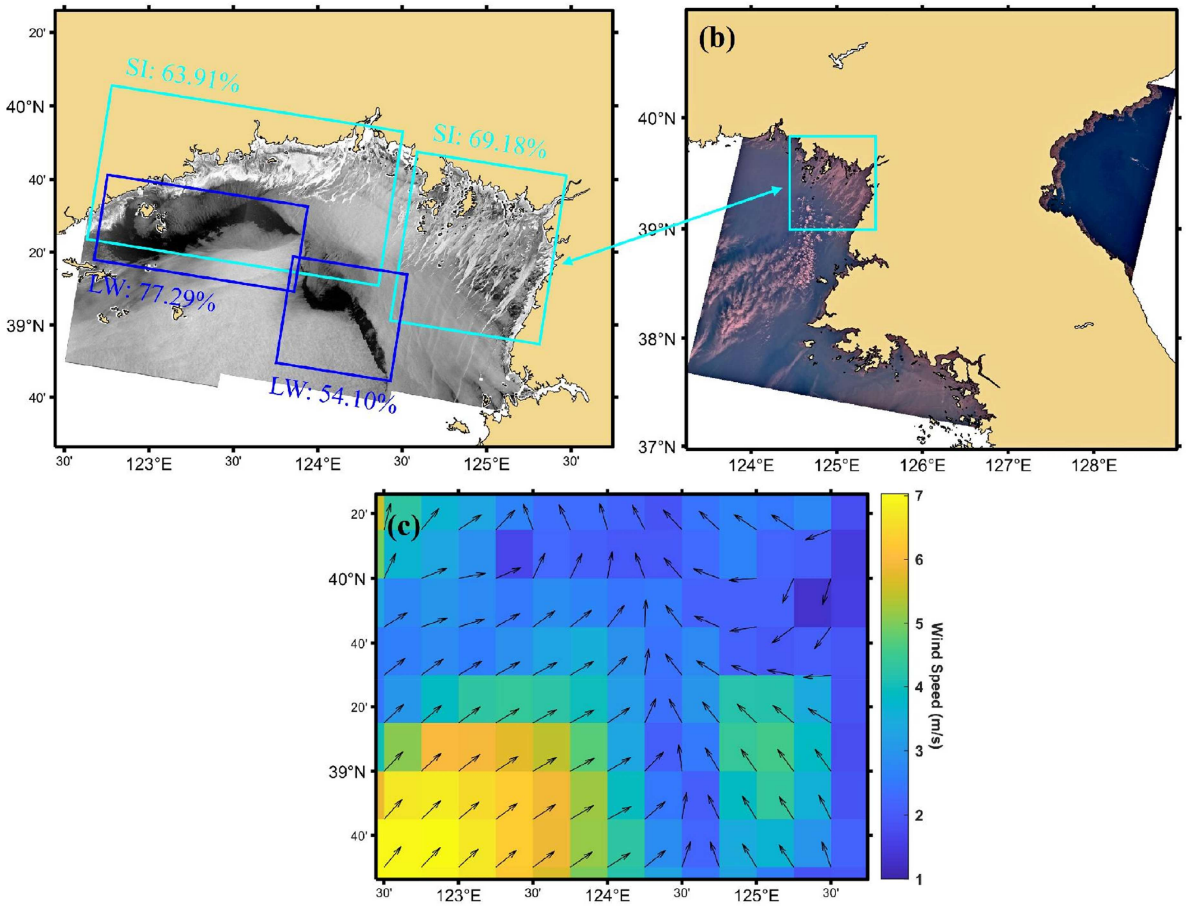


Figure 8. (a) Multi-class dark object detection results obtained using MDODNet from a Sentinel-1B VV-polarisation SAR image acquired in Interferometric Wide (IW) mode at 21:47 UTC on January 18, 2018. Blue and cyan rectangular bounding boxes denote correctly detected low-wind areas (LW) and sea ice (SI), respectively. Numerical values indicate the corresponding detection confidence scores. (b) HJ-1B CCD true-colour composite image acquired at 02:31 UTC on January 19, 2018, covering the area shown in panel (a). (c) ECMWF ERA5 reanalysis wind field at 22:00 UTC on January 18, 2018.

commonly observed in marginal ice zones and transitional freeze–thaw environments. Figure 8b shows an optical image acquired by the HJ-1B satellite at 02:31 UTC on January 19, 2018, covering an adjacent portion of the SAR scene. The cyan rectangle in Figure 8b spatially corresponds to the sea ice region located on the right side of Figure 8a. In the optical image, this region exhibits high-reflectance grey–white features that contrast sharply with the surrounding seawater, confirming the presence of sea ice. The strong spatial consistency between the SAR-detected dark region and the optically bright ice surface further validates the reliability of the MDODNet detection results.

Figure 8c presents ECMWF ERA5 reanalysis wind fields corresponding to the SAR acquisition time. Wind speeds in the vicinity of the low-wind areas detected by MDODNet are below 3 m/s. Additionally, Sharp wind direction discontinuities are observed, forming frontal-like structures. Such discontinuities can generate localised calm zones where surface roughness is reduced, leading to diminished radar backscatter and the appearance of large dark patches in SAR imagery (Johannessen et al. 2012). The strong spatial agreement between the reanalysis wind field and the detected low-wind patches confirms that these features are driven by meteorological conditions rather than being associated with sea ice or oil-related phenomena.

Figure 9a presents the detection results of MDODNet applied to a Sentinel-1 VV-polarisation SAR image acquired in IW mode at 10:56 UTC on July 1, 2017, off the coast of Vietnam. In this scene, the model identifies an extensive region characterised by mixed bright and dark patches, which is classified as a rain

cell (purple bounding box) with an IoU of 0.88. The dark patches exhibit irregular yet relatively smooth boundaries and form a cellular spatial pattern, consistent with typical SAR signatures of rain cells. Such features are commonly associated with rain-induced attenuation and turbulence, which suppress short gravity–capillary waves and reduce radar backscatter (Alpers et al. 2016). Figure 9b shows precipitation data from the GPM acquired at 11:00 UTC on the same day, closely matching the SAR acquisition time. The precipitation field spatially coincides with the dark region detected by MDODNet, confirming the presence of active rainfall. The strong spatial correspondence between the SAR-derived dark patches and the independent precipitation measurements indicates that the observed backscatter reduction is attributable to rain-induced surface and atmospheric effects, rather than oil spills or low-wind phenomena. This case further demonstrates the capability of MDODNet to discriminate meteorologically driven dark signatures from anthropogenic or oceanographic features in complex marine environments.

These three representative cases demonstrate the capability of MDODNet to detect and differentiate multiple categories of dark objects in SAR imagery under complex marine conditions. In the first case, the model successfully distinguished coexisting classes, including biogenic slicks, low-wind areas, and oil spills, within a single scene, highlighting its strong multi-class discrimination capability. When integrated with UNet++, MDODNet further enables accurate pixel-level delineation of oil spill boundaries, facilitating quantitative assessment of spill extent and morphology. The second and third cases further illustrate the robustness of MDODNet in distinguishing oil spills from various look-alike phenomena, including sea ice, rain cells, low-wind areas, and biogenic slicks, which are common sources of false alarms in conventional dark-spot approaches. These results indicate that MDODNet effectively leverages multi-scale representations and contextual information to reduce inter-class confusion in heterogeneous marine environments. Taken together, these case studies validate the robustness and generalisation capability of MDODNet for wide-swath SAR imagery under both single-type and multi-type dark object scenarios. Moreover, the incorporation of multi-source auxiliary datasets, including optical imagery, reanalysis wind fields, and precipitation measurements, provides independent cross-validation, thereby enhancing the credibility of the detection results and highlighting the practical applicability of the framework for real-world marine surveillance. Nevertheless, it is important to acknowledge that the proposed model is not without limitations and remains susceptible to false detections and missed detections. Missed detections are more likely to occur for small-scale dark objects or those with low contrast relative to the surrounding background. In addition, in highly complex scenarios where multiple dark object categories overlap or intersect, ambiguous boundaries may lead to both false detections and missed detections.

4.5. Transferability and generalisation test of MDODNet

To qualitatively assess the generalisation capability of MDODNet across different polarisation configurations, imaging modes, and SAR platforms, preliminary portability experiments were conducted. Although

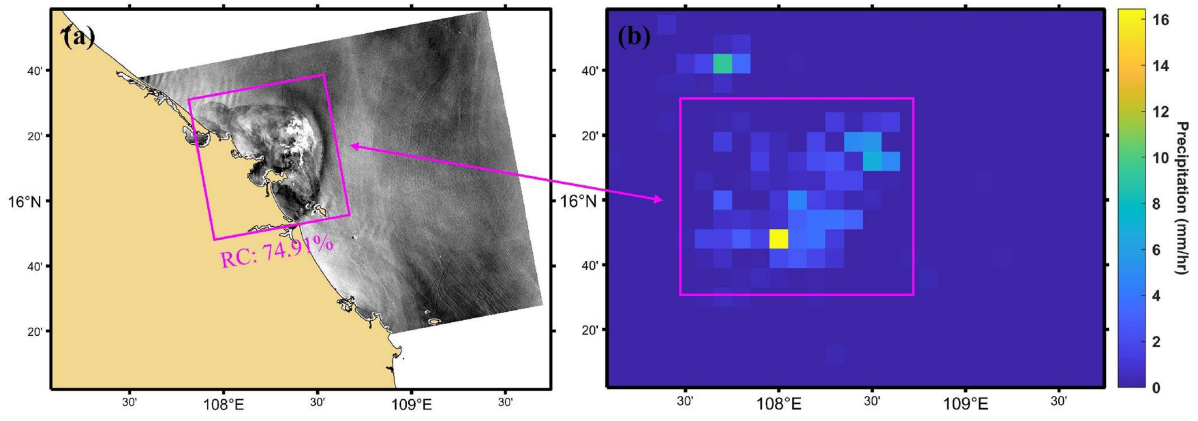


Figure 9. (a) Multi-class dark object detection result obtained using MDODNet from a Sentinel-1A VV-polarisation SAR image acquired in Interferometric Wide (IW) mode at 10:56 UTC on July 1, 2017. The purple bounding box denotes the correctly detected rain cell (RC), and the numerical value indicates the corresponding detection confidence score. (b) Rain rate derived from Global Precipitation Measurement (GPM) data at 11:00 UTC on July 1, 2017.

the training dataset consists exclusively of Sentinel-1 VV-polarisation images acquired in IW mode, practical marine remote sensing applications frequently involve heterogeneous SAR data from multiple sensors with varying acquisition geometries and polarisation schemes. Evaluating model performance under such conditions is therefore essential for assessing robustness and real-world deployment potential. To this end, three representative SAR scenes were selected for cross-platform validation: (1) Sentinel-1 IW mode HH-polarisation imagery, (2) Sentinel-1 EW mode VV-polarisation imagery, and (3) RADARSAT-2 ScanSAR Narrow mode VV-polarisation imagery. Each scene contains characteristic dark patches, including oil slicks and similar oceanic features. The pre-trained MDODNet and UNet++ models were directly applied to these datasets for inference without any transfer learning, fine-tuning, or parameter adjustment. This zero-shot evaluation strategy enables assessment of the intrinsic generalisation capability of the model across variations in polarisation, swath width, spatial resolution, and sensor-specific backscatter characteristics.

Figure 10 presents the detection results of MDODNet applied to a Sentinel-1 HH-polarisation SAR scene acquired in IW mode over the Labrador Sea at 10:11 UTC on June 24, 2020. Despite being trained exclusively on Sentinel-1 VV-polarisation data, the model successfully identified one biogenic slick (green bounding box), two low-wind areas (blue bounding boxes), and one sea ice region (cyan bounding box), demonstrating promising generalisation capability across different polarisation configurations. In late June, the Labrador Sea typically undergoes the seasonal transition from ice-covered to open water conditions. During this period, melt ponds develop across the ice surface, and ice floe boundaries become progressively smoother as melting advances. These processes reduce surface roughness at the radar wavelength scales and suppress Bragg scattering, resulting in diminished radar backscatter that appears as dark regions in SAR imagery. The cyan bounding box in Figure 10 corresponds to such melt-affected sea ice, whose morphological characteristics are consistent with late-season ice conditions. From May to September, enhanced marine biological activity along the Newfoundland and Labrador coasts frequently produces surface-active films that dampen short gravity–capillary waves, giving rise to biogenic slick signatures in SAR imagery. These slicks often coexist spatially with low-wind areas, both appearing as dark patches but arising from distinct physical mechanisms. The results indicate that MDODNet effectively differentiates between these look-alike phenomena without misclassifying them as oil spills, demonstrating robust feature discrimination across diverse environmental conditions and polarisation settings.

Figure 11a presents the detection results of MDODNet applied to a Sentinel-1 VV-polarisation SAR image acquired in EW mode over the Algerian coast at 17:37 UTC on October 21, 2014. Despite differences in swath width and acquisition configuration relative to the training data, the model successfully identified one biogenic slick (green bounding box), three low-wind areas (blue bounding boxes), and two oil spills (red bounding boxes), demonstrating stable performance under alternative imaging modes. The detected

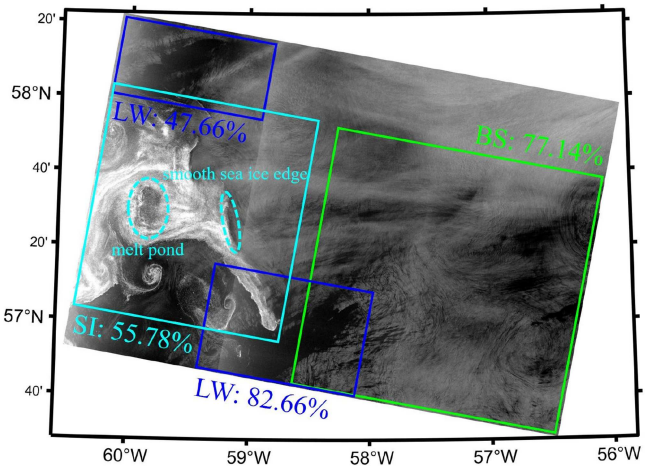


Figure 10. Multi-class dark object detection results obtained using MDODNet from a Sentinel-1B HH-polarisation image acquired in Interferometric Wide (IW) mode at 10:11 UTC on June 24, 2020. Green, blue, and cyan bounding boxes denote detected biogenic slicks (BS), low-wind area (LW), and sea ice (SI), respectively. Numerical values indicate the corresponding detection confidence scores.

oil spills exhibit elongated, linear morphologies, which are commonly associated with vessel-related discharges or illegal waste-oil dumping events in high-traffic maritime corridors. Such strip-like features typically arise from sustained release along a moving vessel trajectory, producing characteristic narrow slick patterns in SAR imagery. The detected oil spill patches were subsequently provided as input to the UNet++ segmentation model. The resulting segmentation map (Figure 11b) delineates spill boundaries with high spatial precision and yields an estimated total spill area of 24.88 km^2. This case further demonstrates the complementary integration of object-level detection and pixel-level segmentation, enabling both accurate localisation and quantitative assessment of oil spill extent under diverse SAR acquisition conditions.

Figure 12a presents the detection results of MDODNet applied to a RADARSAT-2 VV-polarisation SAR scene acquired in ScanSAR Narrow mode over the Gulf of Mexico at 23:51 UTC on August 20, 2018. Despite differences in sensor platform, imaging mode, and radiometric characteristics relative to the Sentinel-1 training data, the model successfully identifies one low-wind area (blue bounding box), four oil spill patches (red bounding boxes), and six rain cells (purple bounding boxes), demonstrating promising cross-sensor generalisation capability. The Gulf of Mexico is well known for persistent natural hydrocarbon seepage, which frequently generates surface oil slicks detectable in SAR imagery. The four detected oil spill patches exhibit irregular, flocculent morphologies consistent with natural seep signatures, rather than narrow linear patterns typically associated with vessel-related discharges. In addition, multiple rain cell signatures are detected, characterised by spatially heterogeneous dark regions resulting from rainfall-induced backscatter attenuation. The simultaneous presence of meteorological, oceanographic, and pollution-related dark features highlights the complexity of this scene and underscores the importance of reliable multi-class discrimination. The detected oil spill regions are subsequently processed using UNet++ for pixel-level segmentation. The resulting segmentation map (Figure 12b) delineates the spill boundaries with strong spatial continuity and yields an estimated total spill coverage of 60.18 km^2. This case further confirms the portability of the proposed detection–segmentation framework across different SAR platforms and acquisition modes, supporting its potential for operational, large-scale marine environmental monitoring.

These three representative cases provide preliminary evidence that the proposed framework can detect multi-class dark objects and perform oil spill segmentation across different SAR platforms, polarisation configurations, and imaging modes. However, as these results are derived from a limited number of scenes, they should be regarded as an initial qualitative indication of transferability rather than a rigorous

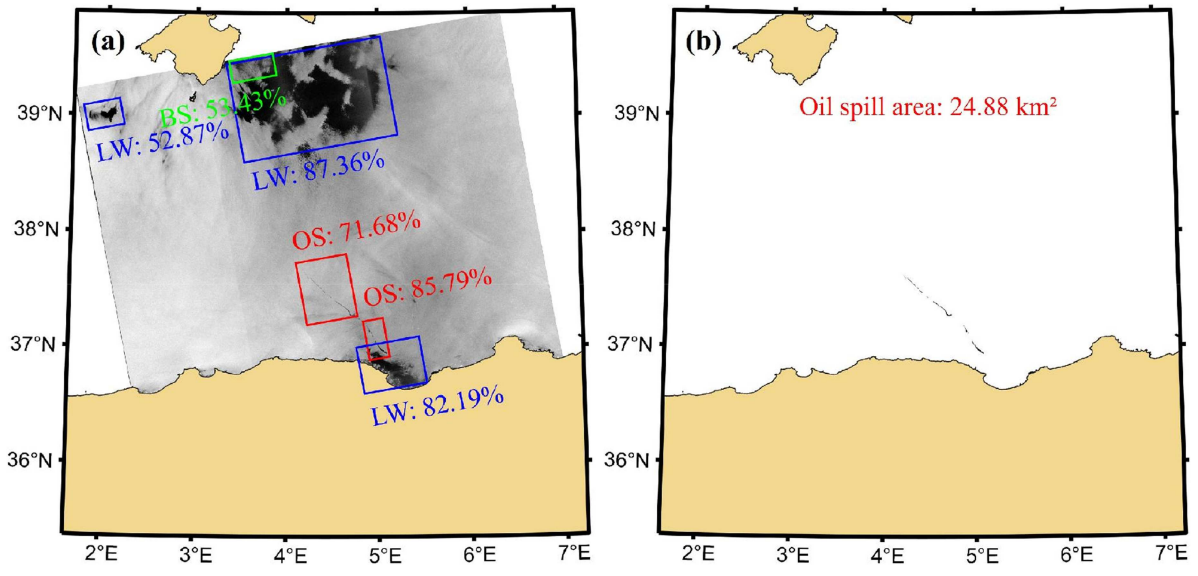


Figure 11. (a) Multi-class dark object detection results obtained using MDODNet from a Sentinel-1A VV-polarisation image acquired in Extra Wide (EW) mode at 17:37 UTC on October 21, 2014. Green, blue, and red bounding boxes denote detected biogenic slicks (BS), low-wind area (LW), and oil spills (OS), respectively. Numerical values indicate the corresponding detection confidence scores. (b) Oil spill segmentation results generated using the UNet++ model. The input patches for UNet++ were derived from the oil spill regions identified by the MDODNet model in panel (a).

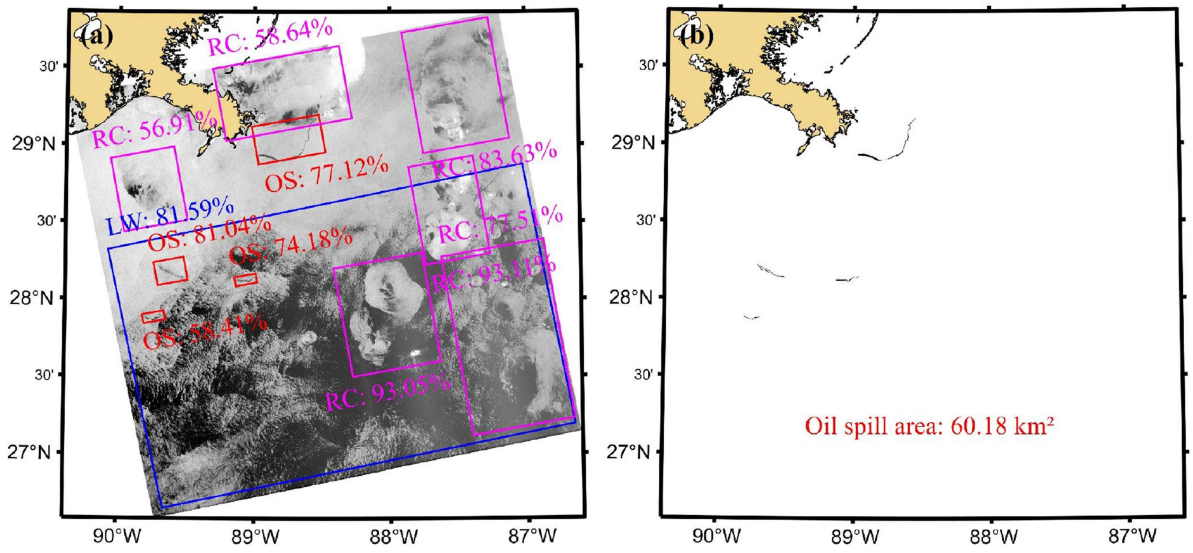


Figure 12. (a) Multi-class dark object detection results obtained using MDODNet from a RADARSAT-2 VV-polarisation image acquired in ScanSAR Narrow mode at 23:51 UTC on August 20, 2018. Blue, red, and purple bounding boxes denote detected low-wind areas (LW), oil spills (OS), and rain cells (RC), respectively. Numerical values indicate the corresponding detection confidence scores for each dark object. (b) Oil spill segmentation results generated using the UNet++ model. The input patches for UNet++ were derived from the oil spill regions identified by the MDODNet model in panel (a).

demonstration of generalisation capability. Achieving reliable operational deployment across diverse SAR systems requires a more comprehensive evaluation. Future work should incorporate a substantially larger and more diverse dataset encompassing a wide range of sensor platforms, frequency bands (e.g. C-, L-, and X-band), polarisation configurations, incidence angles, and imaging modes. In addition, systematic generalisation assessments and the integration of domain adaptation strategies should be considered to further enhance the robustness and scalability of the framework under heterogeneous SAR observation conditions.

5. Conclusion

This study aims to address the long-standing challenge of automatically detecting and discriminating marine oil spills from visually similar look-alike phenomena in wide-swath SAR imagery. To support robust model development, we constructed a validated multi-class dataset comprising 2,192 Sentinel-1 VV-polarisation images acquired in IW mode. Each image was carefully annotated with five categories of dark objects: biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice. The annotation process was incorporated with multispectral optical imagery, ERA5 reanalysis wind fields, and GPM precipitation data, ensuring both label reliability and environmental diversity for model training and evaluation.

Building upon this dataset, we proposed MDODNet, a dedicated multi-class dark object detection framework. MDODNet integrates an extended CSPDarkNet backbone to enhance multi-scale feature representation, an expanded PAFPN for improved cross-scale feature fusion, and an anchor-based detection head optimised with the Focal-EIoU loss to improve localisation precision and regression stability. Evaluation on the test dataset demonstrates strong detection performance, achieving AP values of 87.04%, 93.28%, 88.82%, 90.04%, and 89.16% for biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice, respectively. For quantitative spill assessment, detected oil spill regions were further refined using UNet++, enabling accurate pixel-level delineation of spill boundaries.

Comprehensive experiments and comparative analyses highlight the effectiveness and robustness of the proposed framework. Fine-grained classification of five distinct dark object categories outperforms simplified strategies that either focused solely on oil spills or merged all look-alike phenomena into a single non-oil class. Ablation studies systematically validate the contributions of each architectural component, while comparisons with the original YOLOv8 baseline demonstrate AP improvements of 3.31%, 3.86%, 6.80%, 6.77%, and 4.43% for biogenic slicks, low-wind areas, oil spills, rain cells, and sea ice, respectively.

Comparisons with the representative benchmark models, including SSD, FCOS, and YOLOX, further confirm that the proposed architectural enhancements enable superior discrimination of multi-class marine dark objects in SAR imagery. Representative case studies demonstrate the capability of MDODNet to distinguish oil spills from meteorological and oceanographic look-alike phenomena under complex environmental conditions. Cross-validation using auxiliary geophysical datasets further strengthens the physical interpretability and credibility of the detection results. In addition, preliminary generalisation experiments across different SAR polarisation configurations and imaging modes, including Sentinel-1 HH-polarisation IW mode, Sentinel-1 VV-polarisation EW mode, and RADARSAT-2 VV-polarisation ScanSAR Narrow mode, demonstrate promising transferability beyond the original training configuration.

Despite these encouraging results, several limitations should be acknowledged. From a computational perspective, the current implementation has not yet been optimised for parallel processing or hardware acceleration, which limits its efficiency when handling large-scale, continuous SAR data streams. The proposed MDODNet provides a practical balance between detection accuracy, computational efficiency, and deployment complexity for SAR oil spill monitoring. Recent Transformer-based architectures, such as Vision Transformer (ViT), Swin Transformer, and DETR-style detectors, have demonstrated strong capabilities in modelling long-range spatial dependencies and global contextual information for remote sensing image analysis. Such capabilities may further improve the discrimination of complex marine dark objects in SAR imagery. However, Transformer-based models generally require larger training datasets, higher memory consumption, and substantially greater computational resources than CNN-based approaches. Consequently, optimised CNN-based frameworks such as MDODNet remain advantageous in data-limited and resource-constrained environments that are commonly encountered in operational marine oil spill monitoring. This favourable balance between detection performance, computational efficiency, and deployment feasibility constitutes an important practical strength of the proposed approach.

Similarly, emerging physics-informed and hybrid physics–data–driven approaches offer promising opportunities to improve model interpretability and robustness. By incorporating prior knowledge of SAR scattering physics, sea surface wave damping mechanisms, and physical mechanisms governing marine oil spills and look-alike phenomena, these approaches may enhance model generalisation under diverse environmental conditions and improve the physical consistency of detection results. Nevertheless, neither of these methodological directions was investigated in the present study, primarily due to limited data availability and computational resources.

Future work will include systematic comparisons with Transformer-based detectors and exploration of physics-informed learning strategies. These approaches may further enhance contextual representation, generalisation capability, and operational robustness. Meanwhile, maintaining the computational efficiency required for near-real-time, large-scale SAR oil spill surveillance will remain an important consideration. Furthermore, although the dataset encompasses diverse environmental conditions, additional validation under complex sea states and multi-sensor observation scenarios is still required to fully assess the robustness and transferability of the proposed framework.

In summary, this study provides both a benchmark multi-class dark object dataset and a robust detection–segmentation framework that advances intelligent marine oil spill monitoring in wide-swath SAR imagery. The integration of MDODNet with UNet++ enables reliable multi-class detection alongside precise oil spill delineation, supporting operational oil spill surveillance and environmental assessment. Future work will focus on optimising computational efficiency through parallelisation and hardware acceleration, exploring unified end-to-end instance segmentation architectures to reduce inference latency, and expanding the dataset to include additional dark object categories, multi-sensor configurations, and complex environmental conditions. A comprehensive evaluation using large-scale, multi-sensor SAR datasets will be essential to fully establish the framework's operational readiness.

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Author contributions

CRediT: **Xudong Huang:** Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; **Biao Zhang:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing; **William Perrie:** Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Changjing Xu:** Data curation, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing; **Jinyu Sheng:** Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing.

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Data availability statement

The Sentinel-1 SAR annotated multi-class ocean dark object dataset used in this study is publicly available via the Open Science Framework (OSF) at https://doi.org/10.17605/OSF.IO/VNMKJ (Huang et al. 2026). The dataset is released under a Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license. Users are encouraged to use the dataset for research purposes, provided that an appropriate citation is given to both the dataset and this paper. The satellite and auxiliary data used in this study are publicly available. Sentinel-1A/B SAR data are available from the Alaska Satellite Facility Distributed Active Archive Centre (ASF DAAC) (https://asf.alaska.edu) (Torres et al. 2012). ERA5 reanalysis data are obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) Climate Data Store (CDS) (https://cds.climate.copernicus.eu) (Hersbach et al. 2020). Global Precipitation Measurement (GPM) data are obtained from the National Aeronautics and Space Administration (NASA) (https://gpm.nasa.gov) (Hou et al. 2014). HY-1C/D data are provided by the National Satellite Ocean Application Service (NSOAS) (https://osdds.nsoas.org.cn/DataRetrieval) (Jiang et al. 2023). GF-1/6 and HJ-1/2 data are obtained from the China Platform of Earth Observation System (https://www.cpeos.org.cn/) (Chen et al. 2022; Wang et al. 2010). Landsat-7/8/9 data are obtained from the United States Geological Survey (USGS) (https://earthexplorer.usgs.gov/) (Irons, Dwyer, and Barsi 2012).

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